Carbon Taxes and Clean Technology Transition in an International Environment

Ridhwaan Korimbocus[1]
[1]University of Bristol
ridhwaan.korimbocus@bristol.ac.uk

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Abstract

I study optimal fiscal policy to address climate change in a two-country model with international trade, endogenous technological change, and competition between clean and dirty technologies. Carbon taxation affects both the technology used in current production and the direction of future R&D. Quantitatively, the welfare maximising tax rate is 9% under cooperation between two symmetric countries calibrated to the US and the EU, and in the asymmetric calibration, the cooperative optimal tax rates are 6% in the US and 30% in China, generating significant environmental benefits from lower emissions, but at the cost of a lower average output growth rate. I show that unilateral carbon taxes create carbon leakage by shifting production toward the untaxed jurisdiction, weaken clean R&D incentives and slow the green technological transition. The central contribution is to show that international competition creates strong free-riding incentives and affects the effectiveness of climate policies through both a static production-relocation channel and a dynamic innovation channel.

1 Introduction

International climate policies have long faced the problem of coordinating national actions in the presence of a global environmental externality. Agreements such as the Kyoto Protocol and the Paris Agreement attempted to overcome strong free-riding incentives, and coordinate emissions reductions across countries. At the same time, environmental policies can accelerate the transition from dirty to clean technologies by changing both current and future production incentives through R&D (Acemoglu et al., 2012; Aghion et al., 2016). Yet climate policies are typically implemented at the national level, while firms compete in international markets and the benefits of lower emissions are global. This creates an important tension. A carbon tax can make clean technologies more profitable within the country that implements it, but it may simultaneously shift production toward untaxed foreign jurisdictions. Dechezleprêtre and Sato (2017) summarise the empirical evidence that differences in environmental regulations across countries have negative effects on trade, employment and short-run productivity while inducing domestic clean innovation, although the resulting benefits do not appear sufficient to fully offset the costs.

This paper studies how international coordination changes the effectiveness of carbon taxation when technological change is endogenous. I develop a dynamic general equilibrium model with directed technical change à la Acemoglu et al. (2016) comprising two countries, international trade and a global environmental externality. In each country, a continuum of intermediate goods can be produced using either a clean or a dirty technology. Dirty production uses labour and an exhaustible resource that generates carbon emissions, whereas clean production uses labour alone. In each sector, clean and dirty firms compete in a quality-ladder innovation race. Firms invest in R&D by hiring scientists to improve their technologies, and successful innovation changes the productivity gap between clean and dirty producers. Emissions accumulate in a common atmospheric carbon stock and increase global temperature, which enters household welfare as in long-standing DICE models (Nordhaus, 1992, 2008; Barrage, 2020). The model therefore combines three margins that are central to the analysis, namely, the choice between clean and dirty technology, endogenous technological change through R&D, and international competition between producers facing different national climate policies.

Carbon taxation affects the economy through both a static and a dynamic channel. First, by increasing the marginal cost of dirty production, the tax changes which technology operates in a given sector. A sufficiently high tax can induce a switch from dirty to clean production even when the dirty firm is more productive. Second, this change in relative

profitability affects firms' incentives to invest in R&D. By increasing the value of clean production relative to dirty production, carbon taxation redirects research effort toward clean technologies and changes the subsequent evolution of the productivity distribution.

To establish the benchmark mechanisms, I first calibrate the model to a closed economy, replicating key features of the US, including emissions, R&D intensity, output growth, and the initial distribution of clean and dirty technology gaps. Under a time-invariant policy introduced in 2025, the welfare-maximising carbon tax rate imposed on the costs of dirty firms, that takes into account the welfare costs of climate change, is 12%. The policy introduces an additional mechanism that speeds up the green transition in addition to the underlying force of exhaustible resource getting more expensive. Relative to the no-tax economy, the share of dirty firms falls sharply and approximately 40 percentage points of R&D activity shifts from dirty toward clean firms in 2025. The strength of this response depends on the initial distribution of technology gaps. When many sectors are close to the technology switch threshold, a relatively small change in policy can generate large changes in the technology composition. Emissions decline through two margins. First, dirty firms use fewer fossil inputs, and second, production progressively moves from dirty toward clean technologies. However, this government intervention comes at a cost. Average annual GDP growth falls by 0.4 percentage points under optimal policy compared to laissez-faire, reflecting that, as a result of the tax, it is not necessarily the most productive technology that operates.

The closed economy results abstract from an important real-world angle which is the strategic interactions between countries in setting climate policies. Recent work by Farrokhi and Lashkaripour (2025) emphasised that open-economy climate policies must account for how national interventions changes the international allocation of production and emissions. Therefore, I extend the quantitative exercise to a two-country setting, and the results change substantially once international trade is allowed for. When both countries impose the same tax, dirty producers face similar changes in production costs across markets and the tax can redirect both production and innovation toward clean technologies. However, when the policy is unilateral, domestic dirty production becomes more expensive relative not only to domestic clean firms but also to untaxed foreign dirty producers. Demand can therefore shift abroad rather than toward domestic clean production, generating the conventional carbon leakage channel through production relocation. The dynamic effect of the policy is also weakened by reducing the profitability of clean innovation and R&D. The central question of the paper is therefore whether the directed-technical-change benefits of carbon taxation survive once firms compete internationally against producers that are not subject to the same environmental policy.

First, I consider two symmetric countries calibrated to match empirical moments from the US and the EU. The globally optimal coordinated carbon tax is 9% in each country.[1] To isolate the importance of policy coordination, I introduce the same 9% tax unilaterally in one country while the other remains untaxed. Although the tax increases the relative attractiveness of clean technologies within the tax jurisdiction, it also makes production less competitive relative to its untaxed counterpart. Production shifts toward the untaxed country, significantly reducing the effectiveness of the policy. Carbon leakage implies that any reduction in emissions in the tax jurisdiction is offset by higher emissions in the other country.

International competition also affects innovation incentives. Under the globally coordinated tax, the share of R&D directed toward clean technologies is higher relative to unilateral taxation. In 2025, this share is 20 percentage points higher and stays higher throughout until 2100 when it is 10 percentage points higher. International competition generates not only leakage of current emissions but also a leakage of innovation incentives. Production relocating to the untaxed jurisdiction reduces the expected return to clean R&D and weakens the long-run technological response to environmental policies.

I then introduce cross-country asymmetry by calibrating the model to the US and China. The two economies differ in labour productivity, carbon intensity, emissions growth and R&D intensity. Under cooperation, the welfare-maximising time-invariant taxes are 6% in the US and 30% in China. The difference reflects the substantially different economic and emissions structures of the two countries. Under these policies, emissions fall by approximately 3 gigatonnes in the US and 7 gigatonnes in China, and the increase in global temperature by 2100 is substantially reduced relative to the no-tax baseline.

These forces have direct implications for strategic policy setting. I characterise carbon taxation as a one-shot simultaneous-move game in which each country chooses its tax to maximise domestic welfare, taking the other country's policy as given. In the calibrated model, there is no positive welfare-improving unilateral tax. A country imposing a carbon tax bears the domestic costs associated with lower output and reduced competitiveness, while much of the environmental benefit is either offset by foreign emissions or shared globally. The foreign country, in contrast, benefits from the relocation of production without bearing the cost of the policy. This creates a wedge between the cooperative policy and the individually optimal unilateral tax rate.

[1] The closed-economy and symmetric-countries exercises are calibrated separately to different empirical moments for the US and US+EU, respectively. The difference between the 12% closed-economy optimum and the 9% coordinated optimum therefore reflects both the change in economic environment including international trade frictions and differences in calibration.

Finally, I relax the assumption of time-invariant policies. In the symmetric economy, the optimal time-varying tax is hump-shaped. It begins at 11% in 2025, rising to a peak of approximately 13% over the following two decades, and then gradually declining as the clean transition proceeds. Once clean technology becomes sufficiently dominant, the role of taxation in redirecting production and innovation diminishes and the optimal tax eventually approaches zero. The welfare gains, in lifetime-utility-terms, from time-varying policies are 2 percentage points higher relative to the time-invariant case.

Taken together, the results show that the effectiveness of carbon taxes depends crucially on the international environment in which firms produce and innovate. In a closed economy, or under coordinated international policy, carbon taxation can accelerate the green transition by changing both current technology choices and the direction of R&D. Under unilateral policy, however, international competition redirects production toward untaxed jurisdiction and weakens clean innovation incentives. Evaluating climate policy in an open economy therefore requires accounting not only for conventional carbon leakage through trade, but also for its effect on the endogenous evolution of technology.

Related Literature

This paper contributes to three strands of literature. First, it relates to the literature on environmental policy and directed technical change. Acemoglu et al. (2012) develop a growth model in which innovation can be directed toward clean or dirty technologies and show that environmental policies can redirect technical change toward clean production. Acemoglu et al. (2016) introduce a quality-ladder structure with clean and dirty firms competing within product lines, while Aghion et al. (2016) provide empirical evidence from the auto-mobile industry that firms direct innovation toward cleaner technologies when they face higher fuel prices. Related contributions include Bovenberg and Smulders (1995, 1996), Goulder and Schneider (1999), Grimaud et al. (2011), Hassler et al. (2012), Popp (2004), and Van der Zwaan et al. (2002), which study how endogenous technological change interacts with environmental policy, and changes the dynamics and costs of the green transition. Acemoglu and Rafey (2023) further show how expectations about future climate policies can affect current incentives to invest in clean innovation. Building on this literature, I introduce international market competition into a quality-ladder model of directed technical change and study how differences in environmental policies across countries affect technology adoption and the direction of R&D. I show that the green transition is significantly slower when domestic firms compete against untaxed foreign dirty producers. I further show that there is no positive Nash equilibrium tax rate when countries independently set their climate policies.

Second, the paper relates to the literature on optimal climate policy in closed economies. Golosov et al. (2014), Acemoglu et al. (2016), Barrage (2020) and Douenne et al. (2023) study optimal climate policy design in quantitative general equilibrium environments. A separate strand studies environmental policy in open economies and the interaction between climate policy, trade and carbon leakage. Markusen (1975) and Copeland (1996) show how trade instruments can be used in the presence of cross-border pollution, while Kortum and Weisbach (2021) and Weisbach et al. (2023) study unilateral carbon policy and the role of trade instruments in limiting carbon leakage. Farrokhi and Lashkaripour (2025) analyse quantitatively how trade policies can mitigate climate, showing how optimal trade instruments can address carbon leakage and the international relocation of emissions. Relative to these papers, I combine optimal climate policy with endogenous directed technical change in an open economy. This allows international production relocation to affect not only current emissions and competitiveness, but also the future evolution of clean and dirty technologies.

Third, the paper contributes to work studying unilateral environmental policy in open economies with endogenous technological change. Hémous (2016) develops a two-country trade model with directed technical change and shows that unilateral carbon taxation can increase dirty innovation abroad and may fail to achieve sustainable growth, while a combination of clean research subsidies and trade policy can implement a sustainable path. Van den Bijgaart (2017) studies whether unilateral policy can redirect global innovation in a two-country directed technical change model and finds that individual economies such as the US or EU may be too small to shift global technological change, whereas sufficiently large international coalitions can do so.

My paper differs along the following dimensions. First, I introduce international policy interactions into the quality-ladder framework of Acemoglu et al. (2016). Second, whereas Hémous (2016) and Van den Bijgaart (2017) focus primarily on the ability of unilateral policies to achieve sustainable growth, I characterise welfare-maximising carbon taxes under international cooperation and non-cooperation, including the Nash equilibrium. Third, I calibrate the model to symmetric and asymmetric economies to quantify how international competition affects technology adoption, the allocation of R&D, emissions and the effectiveness of carbon taxation. The central quantitative result is that international integration weakens the clean-innovation response to unilateral carbon taxation. When domestic firms compete against untaxed foreign dirty producers, production shifts abroad and the expected return to clean innovation falls. Thus, carbon leakage operates not only through the location of current production, but also through the endogenous evolution of technology.

The rest of the paper proceeds as follows. Section 2 describes the model and section 3 defines the equilibrium. Section 4 explains the calibration strategy, and sections 5 to 7 present the results. Finally, section 8 concludes.

2 Model

The model extends that of Acemoglu et al. (2016) to a two-country dynamic general equilibrium setup in which final output is produced by combining intermediate goods produced using either a clean or a dirty technology, represented by a quality ladder, with R&D that can be directed toward either technology.

2.1 Preferences

Consider an infinite-horizon world economy in discrete time composed of two countries, Home and Foreign, denoted by $m \in \{H, F\}$. Asterisks are used to indicate variables of the Foreign country. In each country, there is a representative household having additively separable preferences over consumption c_t and temperature change T_t. The lifetime utility of Home household is given by

$$W_H \;=\; \sum_{t=0}^{\infty} \beta^t \big[u(c_t) \;+\; v(T_t)\big] \tag{1}$$

where β is the discount rate. The household includes unskilled workers, with measure one, and "scientists", with measure L^s, who are employed in R&D activities. All workers supply one unit of labour inelastically. The household owns all domestic firms. It maximises lifetime utility subject to the following budget constraint

$$c_t \;\leq\; w_t^u \;+\; w_t^s L^s \;+\; \Pi_t \;+\; B_t \tag{2}$$

where w_t^u and w_t^s are the wage rates of unskilled and R&D workers, Π_t is total firm profits and B_t is the net lump-sum tax or transfer from the government. The lifetime utility for the representative Foreign household is given by

$$W_F \;=\; \sum_{t=0}^{\infty} \beta^t \big[u(c_t^*) \;+\; v(T_t)\big] \tag{3}$$

The corresponding Foreign budget constraint is

$$c_t^* \;\leq\; (w_t^u)^* \;+\; (w_t^s)^*(L^s)^* \;+\; \Pi_t^* \;+\; B_t^* \tag{4}$$

2.2 Final Good

There is a homogenous numeraire final good produced in both countries by combining Home and Foreign intermediates $j \in [0,1]$ with an elasticity of substitution equal to one. Each j is combined as follows

$$x_j^{\epsilon} \hat{x}_j^{1-\epsilon}$$

where x_j denotes the demand for Home good j, $\hat{x}_j$ denotes the demand for Foreign good j and $\epsilon \in [0,1]$ is the expenditure share on the domestic good.[2] The Home production function is

$$Y_t = \prod_{j \in (0,1)} \left(x_{j,t}^{\epsilon} \hat{x}_{j,t}^{1-\epsilon} \right)^{dj} = \exp \left\{ \int_0^1 \left(\epsilon \ln x_{j,t} + (1-\epsilon) \ln \hat{x}_{j,t} \right) \, dj \right\} \tag{5}$$

The corresponding Foreign production function is

$$Y_t^* = \prod_{j \in (0,1)} \left((\hat{x}_{j,t}^*)^{\epsilon} (x_{j,t}^*)^{1-\epsilon} \right)^{dj} = \exp \left\{ \int_0^1 \left(\epsilon \ln \hat{x}_{j,t}^* + (1-\epsilon) \ln x_{j,t}^* \right) \, dj \right\} \tag{6}$$

2.3 Intermediate Goods

Intermediate goods $j \in [0,1]$ are produced with either a clean or a dirty technology, denoted by $k = \{c, d\}$, respectively. Each sector j is represented by a dynamic contest between exactly two firms: one specialising in dirty technology research and the other specialising in clean technology research. At any point in time, there is one firm operating in each sector which I denote as $i = l$ (leader) and the other firm denoted as $i = f$ (follower) does not produce. Denoting intermediate goods production as $y_{j,t}^k$, the production function for a firm using clean technology is

$$y_{j,t}^c = \theta_{j,t}^c l_{j,t}^c \tag{7}$$

where $l_{j,t}^c$ is unskilled labour choice and $\theta_{j,t}^c$ is labour productivity of a clean firm in product line j. The production function for dirty technology is similar, but also includes the use of an exhaustible resource $e_{j,t}$

$$y_{j,t}^d = \theta_{j,t}^d (l_{j,t}^d)^{1-\gamma} e_{j,t}^{\gamma} \tag{8}$$

[2] If $\epsilon = 1$, the model is that of a closed economy with global environmental externality.

The cost minimisation problem of firms using dirty technology implies

$$e_{j,t} \;=\; \Big(\frac{\gamma}{1-\gamma}\frac{w_t^u}{p_t^e}\Big) l_{j,t}^d \tag{9}$$

Combining (9) with (8) gives a linear relationship between output and labour for dirty firms

$$y_{j,t}^d \;=\; \theta_{j,t}^d \Big(\frac{\gamma}{1-\gamma}\frac{w_t^u}{p_t^e}\Big)^{\gamma} l_{j,t}^d \tag{10}$$

2.4 Exhaustible Resource

There is one exhaustible resource demanded by both countries, exogenously produced at zero marginal cost. The stock of the resource R evolves according to the law of motion

$$R_{t+1} \;=\; R_t \;-\; q_t^e \tag{11}$$

where q_t^e is the optimal extraction at time t which satisfies

$$q_t^e \;=\; \int_0^1 e_{j,t}\; dj \;+\; \int_0^1 e_{j,t}^*\; dj \tag{12}$$

Lemma 1. *The price of the exhaustible resource p_t^e follows Hotelling's Rule and grows at the exogenous world interest rate r_t^w, and the price path can be written as*

$$p_t^e \;=\; \left(\prod_{s=1}^{t}(1+r_s^w)\right) p_0^e \tag{13}$$

Proof. See Appendix. □

This is the standard no-arbitrage condition for an exhaustible resource. The resource owner must be indifferent between extracting and selling one unit today at price p_t^e, or leaving it underground to sell in the future at a higher price of $(1+r_{t+1}^w)p_{t+1}^e$.

2.5 Technology choices

In each sector j, the firm with the most advanced technology within clean or dirty will find it profitable to produce. However, because of taxes on the costs of dirty firms τ_t^d, it is not necessarily the most advanced technology that will be used. The marginal costs by technology are

$$MC_c: \qquad \frac{w_t^u}{\theta_{j,t}^c} \tag{14}$$

$$MC_d : \qquad \frac{(1+\tau_t^d) w_t^u \tilde{P}_t}{\theta_{j,t}^d} \tag{15}$$

where w_t^u is the wage rate of unskilled workers and

$$\tilde{P}_t \;=\; \left[\frac{p_t^e}{w_t^u \gamma}\right]^{\gamma} (1-\gamma)^{\gamma-1} \tag{16}$$

is the normalised price of exhaustible resources at time t.

Lemma 2. *The equilibrium technology choices are*

$$\textit{Clean technology if} \qquad \theta_{j,t}^c \;\geq\; \frac{\theta_{j,t}^d}{(1+\tau_t^d)\tilde{P}_t}$$

$$\textit{Dirty technology if} \qquad \theta_{j,t}^c \;<\; \frac{\theta_{j,t}^d}{(1+\tau_t^d)\tilde{P}_t}$$

Proof. See Appendix. □

The operating technology is determined by relative marginal costs rather than productivity alone.[3] A higher tax rate or exhaustible-resource price raises the cost of dirty technology, allowing a relatively less productive clean firm to operate in equilibrium if the threshold is passed.

2.6 Innovation and the quality ladder

Clean and dirty productivity in each sector $\theta_{j,t}^k$ are represented by a quality ladder that evolve as a result of innovation. Each rung in the quality ladder represents a proportional productivity improvement of $\lambda > 1$. R&D can be directed towards improving clean or dirty productivity, and successful innovation results in either an *incremental innovation* with probability $(1-\alpha)$ or a *breakthrough innovation* with probability α. An incremental innovation results in a productivity increase of one rung in the quality ladder whereas a breakthrough innovation allows the follower to improve by one rung over leader irrespective of the productivity differences. Note that a successful innovation for the leader always result in a productivity improvement of λ. Assume without any loss of generality that initial productivity $\theta_{j,0}^k = 1$ for all k and all j. Thus, labour productivity in sector j at time t can be written as

$$\theta_{j,t}^k \;=\; \lambda^{n_{j,t}^k}$$

[3] In the event of a tie, I assume that the clean firm operates. In the quantitative exercise, there is never a tie given the technology distribution.

where $n^k_{j,t}$ is the effective number of rungs that this technology has climbed since $t = 0$. An *incremental innovation* results in productivity $\theta^k_{j,t+1} = \lambda\theta^k_{j,t}$, whereas a *breakthrough innovation* in technology k that is behind the alternative technology $-k$ results in productivity $\theta^k_{j,t+1} = \lambda\theta^{-k}_{j,t}$, that is, an improvement of λ over the leading alternative technology at time t. In any sector j, the relative productivity of the dirty firm to the clean firm at time t is

$$\frac{\theta^d_{j,t}}{\theta^c_{j,t}} = \lambda^{n^d_{j,t} - n^c_{j,t}} \tag{17}$$

where $n^d_{j,t} - n^c_{j,t}$ is the technology gap between the two firms.

Lemma 3. *The tax-adjusted technology choice threshold is*

$$\begin{aligned} &\textit{Clean technology if} \qquad n^d_{j,t} - n^c_{j,t} \leq m_t \\ &\textit{Dirty technology if} \qquad n^d_{j,t} - n^c_{j,t} > m_t \end{aligned}$$

where

$$m_t = \frac{1}{ln\,\lambda}\left[ln\Big((1+\tau^d_t)\tilde{P}_t\Big)\right]$$

is the price-adjusted policy gap in each sector.

Proof. See Appendix. □

The threshold m_t converts the relative marginal cost condition in Lemma 2 into units of the technology gap. A higher tax rate increases m_t makes clean technology operational for a wider range of productivity gaps, including some sectors in which dirty technology remains more productive.

Corollary 1. *The tax-adjusted technology gap is given by*

$$\lambda^{n^d_{j,t} - n^c_{j,t} - m_t} = \frac{\theta^d_{j,t}}{\theta^c_{j,t}} \frac{1}{1+\tau^d_t} \frac{1}{\tilde{P}_t}$$

Proof. See Appendix. □

The tax-adjusted gap summarises the dirty firm's effective productivity advantage after accounting for the tax and exhaustible-resource costs. An increase in either, reduces this effective advantage, shifting production toward clean technology even when the underlying productivity gap $n^d_{j,t} - n^c_{j,t}$ is unchanged.

2.7 R&D

The probabilities of getting a successful innovation for a leader and follower are

$$p_{j,t}^{k,l}(s_{j,t}^{k,l}) \;=\; 1 \;-\; e^{-s_{j,t}^{k,l}} \qquad\qquad p_{j,t}^{k,f}(s_{j,t}^{k,f}) \;=\; 1 \;-\; e^{-a_f s_{j,t}^{k,f}}$$

where $s_{j,t}^{k,l}$ is the number of scientists hired by the leader sector j in period t and $s_{j,t}^{k,f}$ is the number of scientists hired by the follower, and a_f is a probability parameter for the follower. The value functions showing the innovation incentives for a dirty firm and a clean firm in sector j at time t are given by (18) and (19), respectively

$$\begin{aligned} V_{j,t}^{d,i}(n_{j,t}^{d} - n_{j,t}^{c} - m_t) \;=\; \max_{s_{j,t}^{d,i}\geq 0} \mathbb{1}_{\{n_{j,t}^{d} - n_{j,t}^{c} > m_t\}}\pi_{j,t}^{d} \;-\; w_t^s s_{j,t}^{d,i} \\ +\; \rho \mathbb{E} V_{j,t+1}^{d,i}(n_{j,t+1}^{d} - n_{j,t+1}^{c} - m_{t+1}) \end{aligned} \tag{18}$$

$$\begin{aligned} V_{j,t}^{c,i}(n_{j,t}^{d} - n_{j,t}^{c} - m_t) \;=\; \max_{s_{j,t}^{c,i}\geq 0} \mathbb{1}_{\{n_{j,t}^{d} - n_{j,t}^{c} \leq m_t\}}\pi_{j,t}^{c} \;-\; w_t^s s_{j,t}^{c,i} \\ +\; \rho \mathbb{E} V_{j,t+1}^{c,i}(n_{j,t+1}^{d} - n_{j,t+1}^{c} - m_{t+1}) \end{aligned} \tag{19}$$

where $n_{j,t+1}^{d}$ and $n_{j,t+1}^{c}$ evolve according to following the law of motion when the leader is a dirty firm

$$n_{j,t+1}^{d} - n_{j,t+1}^{c} \;=\; \begin{cases} n_{j,t}^{d} - n_{j,t}^{c} \text{ with probability } (1 - p_{j,t}^{d,l})(1 - p_{j,t}^{c,f}) + p_{j,t}^{d,l} p_{j,t}^{c,f}(1-\alpha) \\ n_{j,t}^{d} - n_{j,t}^{c} + 1 \text{ with probability } p_{j,t}^{d,l}(1 - p_{j,t}^{c,f}) \\ n_{j,t}^{d} - n_{j,t}^{c} - 1 \text{ with probability } (1 - p_{j,t}^{d,l}) p_{j,t}^{c,f}(1-\alpha) \\ -1 \text{ with probability } p_{j,t}^{c,f}\alpha \end{cases} \tag{20}$$

and the following law of motion when the leader is a clean firm

$$n_{j,t+1}^{d} - n_{j,t+1}^{c} \;=\; \begin{cases} n_{j,t}^{d} - n_{j,t}^{c} \text{ with probability } (1 - p_{j,t}^{d,f})(1 - p_{j,t}^{c,l}) + p_{j,t}^{d,f} p_{j,t}^{c,l}(1-\alpha) \\ n_{j,t}^{d} - n_{j,t}^{c} - 1 \text{ with probability } (1 - p_{j,t}^{d,f}) p_{j,t}^{c,l} \\ n_{j,t}^{d} - n_{j,t}^{c} + 1 \text{ with probability } p_{j,t}^{d,f}(1 - p_{j,t}^{c,l})(1-\alpha) \\ +1 \text{ with probability } p_{j,t}^{d,f}\alpha \end{cases} \tag{21}$$

There are four possible path for the technology gap: (i) it stays constant if no firm gets a successful innovation or the leader gets a successful innovation and the follower gets an incremental innovation, (ii) it increases by one step if the leader gets a successful innovation and the follower does not, (iii) it decreases by one step if the follower gets an incremental innovation and the leader does not, and (iv) there is a change in technology if the follower gets a breakthrough innovation.

Let $\mu_{n,t}$ denote the evolution of the distribution of technology gaps with some initial conditions $\left\{\mu_{n,0}\right\}_{n=-\infty}^{\infty}$, where $n = n_j^d - n_j^c$. The probability of a successful incremental innovation at gap n for firms using technology k is denoted by z_n^k. The mass of firms at $n > 1$ evolves as follows

$$\begin{aligned}\mu_{n,t+1} \;=\; & \left[(1-z_{n,t}^d)(1-z_{n,t}^c) \;+\; z_{n,t}^d z_{n,t}^c(1-\alpha)\right]\mu_{n,t} \;+\; (1-z_{n+1,t}^d)z_{n+1,t}^c(1-\alpha)\mu_{n+1,t} \\ & +\; z_{n-1,t}^d(1-z_{n-1,t}^c)\mu_{n-1,t}\end{aligned} \tag{22}$$

The mass of firms with gap n at time $t+1$ includes sectors that remain at gap n, inflows from gap $n+1$ when only the clean follower gets a successful incremental innovation, and inflows from $n-1$ when only the dirty leader gets a successful innovation. The mass of firms at $n = 1$, $n = 0$, $n = -1$ and $n < -1$ evolves, respectively, as

$$\begin{aligned}\mu_{1,t+1} \;=\; & \left[(1-z_{1,t}^d)(1-z_{1,t}^c) \;+\; z_{1,t}^d z_{1,t}^c(1-\alpha)\right]\mu_{1,t} \;+\; (1-z_{2,t}^d)z_{2,t}^c(1-\alpha)\mu_{2,t} \\ & +\; z_{0,t}^d(1-z_{0,t}^c)\mu_{0,t} \;+\; \alpha\sum_{n<0} z_{n,t}^d\mu_{n,t}\end{aligned} \tag{23}$$

$$\begin{aligned}\mu_{0,t+1} \;=\; & \left[(1-z_{0,t}^d)(1-z_{0,t}^c) \;+\; z_{0,t}^d z_{0,t}^c(1-\alpha)\right]\mu_{0,t} \;+\; (1-z_{1,t}^d)z_{1,t}^c\mu_{1,t} \\ & +\; z_{-1,t}^d(1-z_{-1,t}^c)(1-\alpha)\mu_{-1,t}\end{aligned} \tag{24}$$

$$\begin{aligned}\mu_{-1,t+1} \;=\; & \left[(1-z_{-1,t}^d)(1-z_{-1,t}^c) \;+\; z_{-1,t}^d z_{-1,t}^c(1-\alpha)\right]\mu_{-1,t} \;+\; (1-z_{0,t}^d)z_{0,t}^c\mu_{0,t} \\ & +\; z_{-2,t}^d(1-z_{-2,t}^c)(1-\alpha)\mu_{-2,t} \;+\; \alpha\sum_{n>0} z_{n,t}^c\mu_{n,t}\end{aligned} \tag{25}$$

$$\begin{aligned}\mu_{n,t+1} \;=\; \big[(1-z^d_{n,t})(1-z^c_{n,t}) \;+\; z^d_{n,t}z^c_{n,t}(1-\alpha)\big]\mu_{n,t} \;+\; (1-z^d_{n+1,t})z^c_{n+1,t}\mu_{n+1,t} \\ +\; z^d_{n-1,t}(1-z^c_{n-1,t})(1-\alpha)\mu_{n-1,t}\end{aligned} \tag{26}$$

2.8 Carbon cycle

Dirty production creates κ units of carbon per unit of exhaustible resource and clean intermediate production do not create carbon emissions. The total amount of Home and Foreign carbon emissions at time t is

$$K_t \;=\; \kappa \int_0^1 e_{j,t}\; dj \qquad\qquad K^*_t \;=\; \kappa \int_0^1 (e_{j,t})^*\; dj$$

Following Archer (2005), Golosov et al. (2014) and Acemoglu et al. (2016), the atmospheric carbon concentration Z_t[4] is determined as follows:

$$Z_t \;=\; \sum_{l=0}^{t}(1-\delta_l)(K_{t-l} \;+\; K^*_{t-l}) \tag{27}$$

where δ_l is the decay factor, that is, the amount of carbon emitted l years ago still remaining in the atmosphere. Finally, the mean temperature change relative to pre-industrial levels is a function of the atmospheric carbon concentration given by $T(Z_t)$.

3 Equilibrium

3.1 Intermediate goods

Solving equation (5), the Home demands of intermediate goods are

$$x_{j,t} \;=\; \frac{\epsilon Y}{p_{j,t}} \qquad\qquad \hat{x}_{j,t} \;=\; \frac{(1-\epsilon)Y}{p^*_{j,t}} \tag{28}$$

Analogously, solving (6), the Foreign demands of intermediate goods are

$$x^*_{j,t} \;=\; \frac{(1-\epsilon)Y^*}{p_{j,t}} \qquad\qquad \hat{x}^*_{j,t} \;=\; \frac{\epsilon Y^*}{p^*_{j,t}} \tag{29}$$

[4]According to Archer (2005) and Golosov et al. (2014), this specification provides a good approximation of the complex dynamics of carbon concentration in the atmosphere.

Market clearing for each j implies that

$$y_{j,t} \;=\; x_{j,t} \;+\; x^*_{j,t} \qquad\qquad y^*_{j,t} \;=\; \hat{x}_{j,t} \;+\; \hat{x}^*_{j,t} \tag{30}$$

The price set by each leader equals the marginal cost of the follower, that is, the maximum possible markup is extracted. When the leading firm is a dirty firm, the equilibrium price and output is

$$p_{j,t} \;=\; MC_c \;=\; \frac{w^u_t}{\theta^c_{j,t}} \qquad\qquad y^d_{j,t} \;=\; \frac{1}{MC_c}\big(\epsilon Y_t + (1-\epsilon)Y^*_t\big) \tag{31}$$

and when the leading firm is a clean firm, the equilibrium price and output is

$$p_{j,t} \;=\; MC_d \;=\; \frac{(1+\tau^d_t)w^u_t \tilde{P}_t}{\theta^d_{j,t}} \qquad\qquad y^c_{j,t} \;=\; \frac{1}{MC_d}\big(\epsilon Y_t + (1-\epsilon)Y^*_t\big) \tag{32}$$

The equilibrium profits are then given by

$$\pi^d_{j,t} \;=\; \left(1 \;-\; \frac{1}{\lambda^{n^d_{j,t}-n^c_{j,t}-m_t}}\right)\big(\epsilon Y_t + (1-\epsilon)Y^*_t\big) \tag{33}$$

$$\pi^c_{j,t} \;=\; \big(1 \;-\; \lambda^{n^d_{j,t}-n^c_{j,t}-m_t}\big)\big(\epsilon Y_t + (1-\epsilon)Y^*_t\big) \tag{34}$$

where $\lambda^{n^d_{j,t}-n^c_{j,t}-m_t}$ is given in Corollary 1.

3.2 Labour market

3.2.1 Scientists

The labour market-clearing condition for scientists equates total demand from all firms in all sectors to the total supply. The Home market-clearing condition is

$$L^s \;=\; \int_0^1 \sum_k \sum_i s^{k,i}_{j,t}\; dj \tag{35}$$

and the Foreign market-clearing condition is

$$(L^s)^* \;=\; \int_0^1 \sum_k \sum_i (s^{k,i}_{j,t})^*\; dj \tag{36}$$

3.2.2 Unskilled workers

From the production function (7), the labour demand of clean firms is

$$l^c_{j,t} \;=\; \frac{y^c_{j,t}}{\theta^c_{j,t}} \;=\; \frac{\lambda^{n^d_{j,t}-n^c_{j,t}-m_t}}{w^u_t}\big(\epsilon Y_t + (1-\epsilon)Y^*_t\big) \tag{37}$$

by substituting (32) and Corollary 1. From the production function (10), the labour demand of dirty firms is

$$l^d_{j,t} \;=\; \frac{y^d_{j,t}}{\theta^d_{j,t}}\Big(\frac{1-\gamma}{\gamma}\frac{p^e_t}{w^u_t}\Big)^{\gamma} \;=\; \frac{1}{(1+\tau^d_t)(w^u_t)^{1-\gamma}\tilde{P}_t\lambda^{n^d_{j,t}-n^c_{j,t}-m_t}}\Big(\frac{1-\gamma}{\gamma}p^e_t\Big)^{\gamma}\big(\epsilon Y_t + (1-\epsilon)Y^*_t\big) \tag{38}$$

by substituting (31) and Corollary 1.

Finally, the market clearing condition for the unskilled labour market is

$$\begin{aligned} 1 = \int_0^1 & \frac{\lambda^{n^d_{j,t}-n^c_{j,t}-m_t}}{w^u_t}\big(\epsilon Y_t + (1-\epsilon)Y^*_t\big)\cdot \mathbb{1}_{\{n^d_{j,t}-n^c_{j,t}\le m_t\}} \\ & + \frac{1}{(1+\tau^d_t)(w^u_t)^{1-\gamma}\tilde{P}_t\lambda^{n^d_{j,t}-n^c_{j,t}-m_t}}\Big(\frac{1-\gamma}{\gamma}p^e_t\Big)^{\gamma}\big(\epsilon Y_t + (1-\epsilon)Y^*_t\big)\cdot \mathbb{1}_{\{n^d_{j,t}-n^c_{j,t}>m_t\}}\, dj \end{aligned} \tag{39}$$

Analogously, the market clearing condition for the Foreign unskilled labour market is

$$\begin{aligned} 1 = \int_0^1 & \frac{\lambda^{(n^d_{j,t})^*-(n^c_{j,t})^*-(m_t)^*}}{(w^u_t)^*}\big((1-\epsilon)Y_t + \epsilon Y^*_t\big)\cdot \mathbb{1}_{\{(n^d_{j,t})^*-(n^c_{j,t})^*\le m^*_t\}} \\ & + \left(\frac{1}{(1+(\tau^d_t)^*)((w^u_t)^*)^{1-\gamma}\tilde{P_t}^*\lambda^{(n^d_{j,t})^*-(n^c_{j,t})^*-(m_t)^*}}\Big(\frac{1-\gamma}{\gamma}p^e_t\Big)^{\gamma}\big((1-\epsilon)Y_t + \epsilon Y^*_t\big)\right) \\ & \cdot \mathbb{1}_{\{(n^d_{j,t})^*-(n^c_{j,t})^*>m^*_t\}}\, dj \end{aligned} \tag{40}$$

3.3 Resource constraint

Given that households are homogenous, do not save and only consume the domestic non-tradable final good, the individual and global resource constraints are

$$Y_t \;=\; c_t \;+\; \int p^e_t e_t\, dj \qquad\qquad Y^*_t \;=\; c^*_t \;+\; \int p^e_t e^*_t\, dj \tag{41}$$

$$\mathbb{Y}_t \;=\; Y_t \;+\; Y_t^* \;=\; c_t \;+\; c_t^* \;+\; \int p_t^e(e_t + e_t^*)\; dj \tag{42}$$

where $\mathbb{Y}_t$ represents world output which equals consumption and total energy expenditure in each country.

Lemma 4. *Aggregate output can be expressed in terms of tax-adjusted labour productivities*

$$Y_t \;=\; \frac{\epsilon\Omega_t^{-1} - (1-\epsilon)(\Omega_t^*)^{-1}}{2\epsilon - 1} \tag{43}$$

$$Y_t^* \;=\; \frac{\epsilon(\Omega_t^*)^{-1} - (1-\epsilon)\Omega_t^{-1}}{2\epsilon - 1} \tag{44}$$

where

$$\Omega_t = \int_0^1 \frac{\lambda^{n_{j,t}^d - n_{j,t}^c - m_t}}{w_t^u} \cdot \mathbb{1}_{\{n_{j,t}^d - n_{j,t}^c \leq m_t\}} + \frac{1}{(1+\tau_t^d)(w_t^u)^{1-\gamma}\tilde{P}_t\lambda^{n_{j,t}^d - n_{j,t}^c - m_t}} \Big(\frac{1-\gamma}{\gamma} p_t^e\Big)^{\gamma} \cdot \mathbb{1}_{\{n_{j,t}^d - n_{j,t}^c > m_t\}}\; dj$$

$$\Omega_t^* = \int_0^1 \frac{\lambda^{(n_{j,t}^d)^* - (n_{j,t}^c)^* - (m_t)^*}}{(w_t^u)^*} \cdot \mathbb{1}_{\{(n_{j,t}^d)^* - (n_{j,t}^c)^* \leq m_t^*\}} + \frac{1}{(1+(\tau_t^d)^*)((w_t^u)^*)^{1-\gamma}\tilde{P}_t^*\lambda^{(n_{j,t}^d)^* - (n_{j,t}^c)^* - (m_t)^*}} \Big(\frac{1-\gamma}{\gamma} p_t^e\Big)^{\gamma} \cdot \mathbb{1}_{\{(n_{j,t}^d)^* - (n_{j,t}^c)^* > m_t^*\}}\; dj$$

Proof. The proof is trivial by combining equations (39) and (40). □

3.4 Government

The government collects taxes from dirty firms and uses lump-sum rebates to balance its budget each period.

$$B_t \;=\; \int_0^1 \frac{\tau_t^d w_t^u \tilde{P}_t}{\theta_{j,t}^d} y_{j,t}^d\; dj \tag{45}$$

Definition 1. ***(World Equilibrium)***

For given initial distributions of technology gaps $\{n_{j,0}^d - n_{j,0}^c\}_{j=0}^1$ *and* $\{(n_{j,t}^d)^* - (n_{j,t}^c)^*\}_{j=0}^1$*, time path of policies* $\{\tau_t^d, (\tau_t^d)^*\}_{t=0}^\infty$*, the initial price of exhaustible resource* p_0^e *and world*

interest rate $\{r_t^w\}_{t=0}^{\infty}$*, a dynamic equilibrium path for the two countries is characterised by a time path of*

$$\Big\{y_{j,t}^c,\ y_{j,t}^d,\ p_{j,t},\ (y_{j,t}^c)^*,\ (y_{j,t}^d)^*,\ p_{j,t}^*,\ \big(n_{j,t}^d - n_{j,t}^c\big),\ \big((n_{j,t}^d)^* - (n_{j,t}^c)^*\big),\ s_{j,t}^{c,i},\ s_{j,t}^{d,i},\ (s_{j,t}^{c,i})^*,\ (s_{j,t}^{d,i})^*,$$
$$w_t^s,\ (w_t^s)^*,\ w_t^u,\ (w_t^u)^*\ e_{j,t},\ e_{j,t}^*,\ Y_t,\ Y_t^*,\ Z_t\Big\}_{t=0}^{\infty}$$

such that

1. $y_{j,t}^c$*,* $y_{j,t}^d$ *and* $p_{j,t}$ *maximise profits according to* (31) *and* (32)*.*
2. $(y_{j,t}^c)^*$*,* $(y_{j,t}^d)^*$ *and* $p_{j,t}^*$ *maximise profits of foreign intermediate firms.*
3. $n_{j,t}^c$ *and* $n_{j,t}^d$ *follow the law of motions* (20) *and* (21)*.*
4. $s_{j,t}^{c,i}$ *and* $s_{j,t}^{d,i}$ *solve* (18) *and* (19)*.*
5. $(s_{j,t}^{c,i})^*$ *and* $(s_{j,t}^{d,i})^*$ *solve the value functions of foreign intermediate firms.*
6. w_t^s *and* $(w_t^s)^*$ *clear the scientists labour market according to* (35) *and* (36)*.*
7. w_t^u *and* $(w_t^u)^*$ *clear the unskilled labour market according to* (39) *and* (40)*.*
8. $e_{j,t}$ *is given by* (9)*.*
9. Y_t *and* Y_t^* *satisfy the resource constraints in* (41)*.*
10. Z_t *is given by* (27)*.*

Definition 2. ***(Home Equilibrium: Closed economy)***

To define an equilibrium for a closed Home economy, set $\epsilon = 1$ *and let only Home emissions enter the carbon cycle, such that* $Z_t = \sum_{l=0}^{t}(1-\delta_l)K_{t-l}$*.*

For a given initial distribution of technology gaps $\{n_{j,0}^d - n_{j,0}^c\}_{j=0}^{1}$*, time path of policies* $\{\tau_t^d\}_{t=0}^{\infty}$*, the initial price of exhaustible resource* p_0^e *and world interest rate* $\{r_t^w\}_{t=0}^{\infty}$ *and , a dynamic equilibrium path is then characterised by a time path of*

$$\Big\{y_{j,t}^c,\ y_{j,t}^d,\ p_{j,t},\ \big(n_{j,t}^d - n_{j,t}^c\big),\ s_{j,t}^{c,i},\ s_{j,t}^{d,i},\ w_t^s,\ w_t^u,\ e_{j,t},\ Y_t,\ Z_t\Big\}_{t=0}^{\infty}$$

4 Optimal Policy

I now turn to characterising optimal policy. A natural benchmark is the tax a global planner would set to maximise joint welfare, internalising the fact that emissions reduction

in one country also benefits the other through the shared carbon stock. Each country chooses a tax rate $\tau_m^d \in [0, \bar{\tau}]$, with to maximise their welfare $W_m(\tau_m^d, \tau_{-m}^d)$. A global planner chooses both tax rates to maximise aggregate welfare

$$(\hat{\tau}_H^d, \hat{\tau}_F^d) \; = \arg \max_{(\tau_H^d, \tau_F^d) \in [0,\bar{\tau}]} \left\{ W_H(\tau_F^d, \tau_F^d) + W_F(\tau_H^d, \tau_F^d) \right\}$$

The first-order condition with respect to τ_m^d states that the planner equates the direct marginal effect of country m's tax on its own welfare with the external marginal effect on the other country's welfare, given by

$$\left. \frac{\partial W_m(\tau_m^d, \tau_{-m}^d)}{\partial \tau_m^d} \right|_{(\hat{\tau}_H^d, \hat{\tau}_F^d)} + \left. \frac{\partial W_{-m}(\tau_m^d, \tau_{-m}^d)}{\partial \tau_m^d} \right|_{(\hat{\tau}_H^d, \hat{\tau}_F^d)} = 0$$

The net external effect combines the climate benefits to the other country and any general equilibrium spillovers through consumption, wages, prices, or profits. The climate component is positive when a higher tax rate lowers global temperature, while the sign of the general equilibrium component is generally ambiguous. Next, the question is whether each country would individually choose the cooperative tax rates of the planner or not.

Proposition 1. ***(The cooperative solution is not a Nash equilibrium)***

Let $(\hat{\tau}_H^d, \hat{\tau}_F^d) \in (0, \bar{\tau})$ denote an interior optimum. Suppose that, at $(\hat{\tau}_H^d, \hat{\tau}_F^d)$, a marginal increase in country m's tax rate has a non-zero net external effect on the other country, that is,

$$\left. \frac{\partial W_{-m}(\tau_m^d, \tau_{-m}^d)}{\partial \tau_m^d} \right|_{(\hat{\tau}_H^d, \hat{\tau}_F^d)} \neq 0$$

Then $(\hat{\tau}_H^d, \hat{\tau}_F^d)$ is not a Nash equilibrium.

Further, suppose that, at the interior cooperative solution , the net external effect of country m's tax is positive

$$\left. \frac{\partial W_{-m}(\tau_m^d, \tau_{-m}^d)}{\partial \tau_m^d} \right|_{(\hat{\tau}_H^d, \hat{\tau}_F^d)} > 0$$

then country m is individually strictly better off reducing its tax rate below the cooperative level. That is, for a sufficiently small $\varepsilon > 0$

$$W_m(\hat{\tau}_m^d - \varepsilon, \hat{\tau}_{-m}^d) > W_m(\hat{\tau}_m^d, \hat{\tau}_{-m}^d).$$

Proof. See Appendix. □

The condition $\partial W_{-m}/\partial \tau_m^d \neq 0$ states that the tax chosen by country m has a spillover effect on the other country's welfare, in either direction. At the cooperative solution$(\hat{\tau}_H^d, \hat{\tau}_F^d)$, the planner's first-order condition accounts for the effects of the tax on both countries' welfare. If country m were to evaluate $\hat{\tau}_m^d$ privately, it would only account for the effect on its own welfare, ignoring the spillover to the other country. Since the planner's condition and the private condition differ by exactly $\partial W_{-m}/\partial \tau_m^d$, they cannot both hold simultaneously, meaning that $\hat{\tau}_m^d$ does not satisfy country m's private first-order condition. The cooperative solution is therefore not individually optimal, irrespective of the sign or magnitude of the externality. When the external effect of country m's tax is positive, part of the reason the planner chooses $\hat{\tau}_m^d$ is that country m's tax benefits country $-m$. However, when choosing its tax unilaterally, country m does not internalise this benefit. Therefore, at the cooperative allocation, the private marginal return to country m's own tax is lower than the social marginal return, and this creates a local incentive for country m to reduce its tax rate below the cooperative level.

5 Calibration

5.1 Closed economy

The Home economy has 11 parameters $\{\beta, L^s, \gamma, \lambda, a_f, \alpha, \rho, \varphi_p, \varphi_0, \varphi, \kappa\}$, the price path of exhaustible resource p_t^e and the initial technology gaps distribution across intermediate sectors $\{n_{j,0}^d - n_{j,0}^c\}_{j=0}^1$ to be determined. I calibrate the model with an initial gap distribution estimated in the year 2005 to match empirical moments in 2015. Each period is taken to be 10 years.

5.1.1 Preferences and Production

The utility function of households is

$$u(c_t) \; + \; v(T_t) \; = \; \frac{c_t^{1-\sigma}}{1-\sigma} \; + \; \frac{(1+T_t^2)^{-(1-\sigma)}}{1-\sigma}$$

where the functional form for $v(T_t)$ follows Barrage (2020). I set $\sigma = 1.5$, in line with the DICE. This formulation of preferences over climate change implies that the risk aversion coefficient for temperature changes is the similar for utility damages and equivalent consumption losses. The quadratic form reflects the convex and accelerating utility damages from climate change. As Nordhaus and Moffat (2017) explain, the quadratic form is a suitable empirical approximation of the damage function and reflects the convex and accelerating utility damages from climate change.

The change in global mean temperature is modelled as follows

$$T_t \;=\; \eta\Big(\psi ln\Big(\frac{Z_t}{\overline{Z}}\Big)\Big)$$

where $\overline{Z}$ is the pre-industrial level carbon concentration in the atmosphere, typically assumed to be 596 gigatonnes. This formulation reflects the well-established logarithmic relationship between carbon concentration and radiative forcing, and the linear relationship between radiative forcing and temperature change.

To calibrate ψ, I use the equilibrium climate sensitivity (ECS) which measures the change in global average temperature following a doubling of the carbon concentration in the atmosphere relative to its pre-industrial levels. The ECS is estimated to be between 1.5°C to 4.5°C, and taking an average of 3°C, $\psi = 3/ln(2) = 4.33$. Finally, η is calibrated to match an initial temperature of 1.3°C over pre-industrial levels in 2005.

The supply of scientists $L^s = 0.055$ is taken from Acemoglu et al. (2016) who estimates this from the share of full-time workers with college education or higher employed working in jobs related to science and engineering. The annual discount rates for households and firms are set to 1.5%, matching DICE and giving a value of $\rho = \beta = (0.985)^{10}$. Finally, following Golosov et al. (2014), I set the share of exhaustible resource in production to $\gamma = 0.04$.

To compute the world interest rate r^w, I use historical oil price data between 1900 and 2005. The 10-year average price increase is 43%, giving an annual price increase of $r^w = 3.6\%$.

5.1.2 Carbon cycle

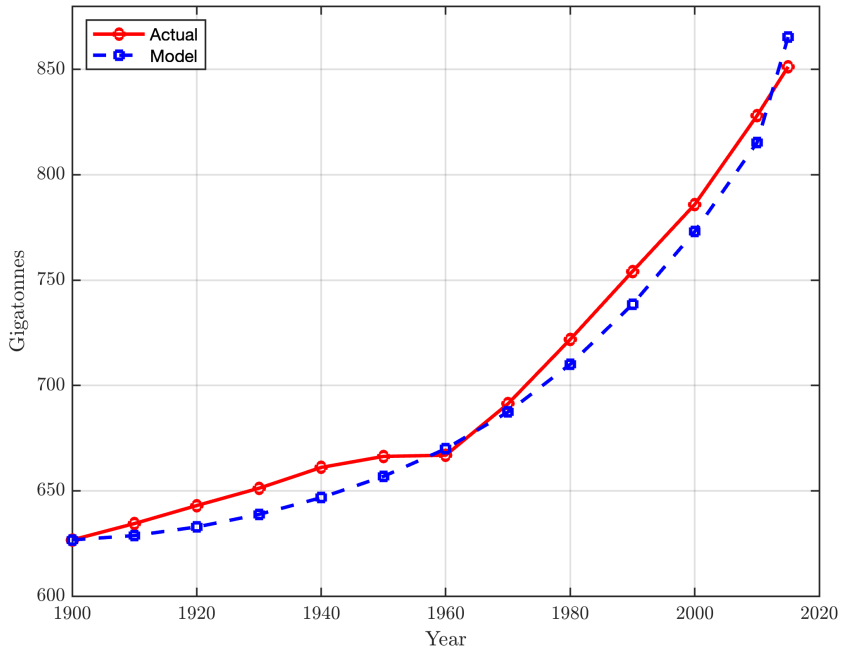
The decay factor of atmospheric carbon δ_l is given by

$$1 - \delta_l \;=\; \varphi_p \;+\; (1 - \varphi_p)\varphi_0\varphi^l \tag{46}$$

where $\varphi_p \in (0, 1)$ denotes the share of carbon emissions that remains permanently in the atmosphere and φ_0 is the fraction of the remaining $(1 - \varphi_p)$ that slowly decays at rate φ over time. I set $\varphi_p = 0.2$ following the estimates in the IPCC (2007) report that about 20% of any emission stays in the atmosphere for thousands of years. I calibrate $\varphi_0 = 0.794$ and $\varphi = 0.758$ to match historical atmospheric carbon concentration from 1990 to 2015, taking the initial stock of carbon Z_{1990} as given. Figure 1 shows the fit of the carbon cycle to the actual data.

Table 1 summarises the parameters of the model.

Figure 1. Atmospheric carbon concentration



Note: To compute the actual data and model results, I combine annual data of carbon dioxide emissions from Global Carbon Budget (2024) – with major processing by Our World in Data, together with atmospheric carbon concentration data from NOAA Global Monitoring Laboratory - Trends in Atmospheric Carbon Dioxide (2025); EPA based on various sources (2022) – with major processing by Our World in Data.

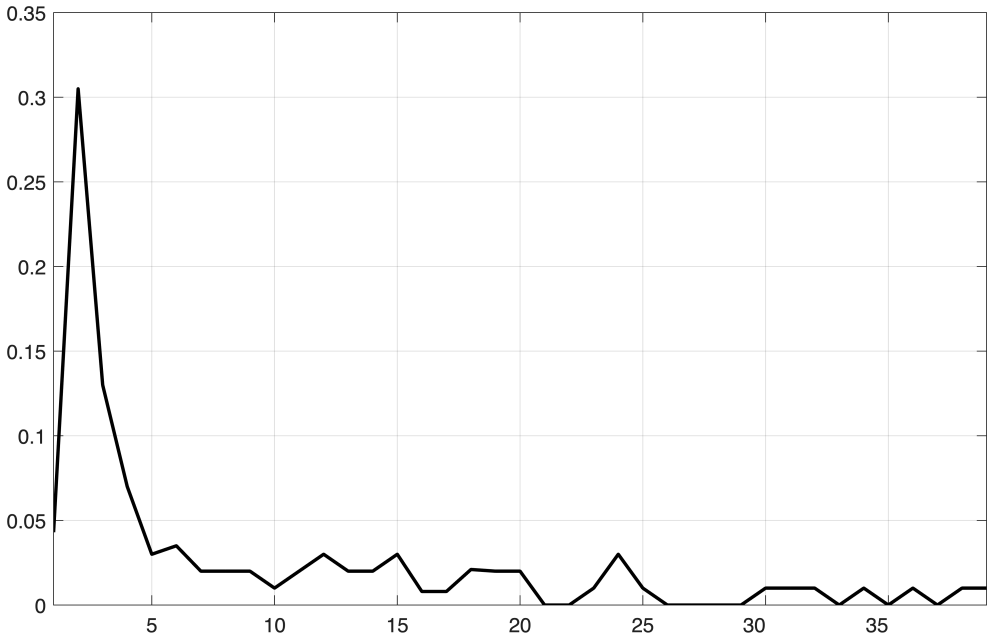
Table 1. Parameters (initial steady state)

Parameter	**Description**	**Value**	**Source or Target**
σ	CRRA coefficient	1.5	Nordhaus (2008)
ψ	Temperature change sensitivity coefficient	4.33	ECS = 3°C
L^s	Measure of scientists	0.055	Acemoglu et al. (2016)
β	Discount rate (Households)	$(0.985)^{10}$	Nordhaus (2008)
ρ	Discount rate (Firms)	$(0.985)^{10}$	Nordhaus (2008)
γ	Exhaustible resource share	0.04	Golosov et al. (2014)
r^w	World interest rate	0.036	Historical oil prices
φ_p	Permanent carbon in the atmosphere	0.2	IPCC (2007)
φ_0	Fraction of non-permanent emission decaying over time	0.794	Historical carbon concentration (Figure 1)
φ	Rate of decay of non-permanent emission	0.758	Historical carbon concentration (Figure 1)
α	Breakthrough probability	0.04	Acemoglu et al. (2016)

5.1.3 Initial productivity gap

I take the initial productivity gaps across sectors from Acemoglu et al. (2016). Using firm-level patent data from the U.S. energy sector between 1975 and 2004, they classify patents as "clean" or "dirty" based on their respective technology codes.[5] Firms are first categorised as clean or dirty in a given year if 25% or more of their energy-related patents are clean and a firm is classified as clean if this threshold is met in at least half of its observed years. Sectors with a predominance of dirty patents are assigned a larger initial gap, while sectors with greater clean patents have a smaller gap, such that the fraction of sectors using clean or dirty technologies matches the observed share of clean and dirty patents in the data.[6] Figure 2 shows the density of initial productivity gaps. It shows that in most sectors, the dirty firm is only a few steps ahead of clean firm. There is a long tail of sectors lines with a large gap between the dirty and clean firms. The fraction of non-zero gaps is 96% percent. The gap is 20 steps or more in 12% of sectors, and 50 steps or more in 2% of sectors.

Figure 2. Initial productivity gap



[5]Clean-energy patents are related to geothermal, hydroelectric, nuclear, solar and wind energy, and conservation and utilisation of energy, while dirty-energy patents are related to the extraction, refinement, or use of fossil fuel–based energy (oil, coal, natural gas, and shale technologies).

[6]The patent counts are converted into discrete innovation steps using a conversion factor calculated as the ratio of the average patent count to the annual innovation rate. Dividing the clean and dirty patent stocks of each sector by the conversion factor yields the initial technological gaps. While Acemoglu et al. (2016) has a very small number of sectors with a negative initial gap, I set them to zero.

5.1.4 Calibrated parameters

The remaining 5 parameters are chosen to match moments from the data. The ladder step size λ is chosen to match the GDP growth rate between 2005 and 2015. The model annualised GDP growth rate is 2.09% compared to the actual growth rate of 3.44%. To calibrate how dirty output translates to emissions, κ is chosen to match actual emissions levels of 5.3 gigatonnes(Gtn) in 2015, with the model generating an emissions level of 5.27Gtn. The initial price of the exhaustible resource p_0^e is chosen to match emissions growth rate between 2005 and 2015. The model annualised emissions growth rate is -0.67% compared to the actual growth rate of -1.24%. The successful innovation probability parameter a_f is calibrated to match an R&D to sales ratio of 3.9% (NCSES Report, 2018) and the model generates a ratio of 3.88%. Finally, I calibrate the parameter η that maps atmospheric carbon concentration to the average temperature change relative to pre-industrial levels, targetting a temperature rise of 2°C, and obtain a rise of 1.7°C in the model. Table 2 shows the fit of the model to the moments.

Table 2. Model fit (Closed economy)

Moment (US)	Data	Model
Actual emissions 2015 (Gtn)	5.3	5.27
Annualised emissions growth rate 2005-2015 (%)	-1.24	-0.67
R&D/Sales ratio 2015 (%)	3.9	3.88
Annualised GDP growth rate 2005-2015 (%)	3.44	2.09
Temperature rise by 2100 (°C)	2	1.7

5.2 Symmetric countries

With two symmetric countries, I take all parameters, the price path of exhaustible resource and the initial technology gaps distribution to be the same in both countries, and calibrate both initial economies to the year 2005. To calibrate the remaining 5 parameters, I choose combined moments of the US and the EU to represent the two symmetric countries. The model generates an annualised GDP growth rate of 2.26%, emissions levels of 8.29Gtn in 2015, annualised emissions growth rate of -0.91%, R&D to sales ratio of 2.96% and temperature rise of 1.75°C, compared to actual GDP growth rate of 2.48%, emissions levels of 8.3Gtn in 2015, emissions growth rate of -1.55%, R&D to sales ratio of 2.97% and temperature rise of 2°C, respectively. Table 3 shows the fit of the model to the moments.

Table 3. Model fit (Symmetric countries)

Moment (US+EU)	Data	Model
Actual emissions 2015 (Gtn)	8.3	8.29
Annualised emissions growth rate 2005-2015 (%)	-1.55	-0.91
R&D/Sales ratio 2015 (%)	2.97	2.96
Annualised GDP growth rate 2005-2015 (%)	2.48	2.26
Temperature rise by 2100 (°C)	2	1.75

5.3 Asymmetric Countries

With two asymmetric countries, I calibrate 4 parameters $\{\kappa, p_0^e, L_s, \lambda\}$ to match country-specific 2015 moments in the US and China, and calibrate η to match the global temperature rise. To introduce asymmetry between countries, I allow the two countries to differ in labour productivity and share of exhaustibles used in production. I parametrise labour productivity using the ratio of US GDP to China GDP as calibration target, and set $\gamma = 0.04$ for the US and $\gamma^* = 0.09$ for China, consistent with the fact that production in China is 2.3 times more carbon-intensive than the US. For the US, the model generates an annualised GDP growth rate of 2.42%, emissions levels of 5.3Gtn in 2015, annualised emissions growth rate of -1.15% and R&D to sales ratio of 3.88%, compared to actual GDP growth rate of 3.44%, emissions levels of 5.3Gtn in 2015, emissions growth rate of -1.24% and R&D to sales ratio of 3.9%. For China, the model generates an annualised GDP growth rate of 9.17%, emissions levels of 9.9Gtn in 2015, annualised emissions growth rate of 5.64% and R&D to sales ratio of 2.01%, compared to actual GDP growth rate of 9.5%, emissions levels of 9.9Gtn in 2015, emissions growth rate of 5.3% and R&D to sales ratio of 2.02%. The model generates a temperature rise of 2.01°C and GDP ratio of 1.54, compared to empirical moments of 2°C and 1.61, respectively. Table 4 shows the fit of the model to the moments.

Table 4. Model fit (Asymmetric countries)

	US		China	
Country-specific moments	Data	Model	Data	Model
Actual emissions 2015 (Gtn)	5.3	5.3	9.9	9.9
Annualised emissions growth rate 2005-2015 (%)	-1.24	-1.15	5.3	5.64
R&D/Sales ratio 2015 (%)	3.9	3.88	2.02	2.01
Annualised GDP growth rate 2005-2015 (%)	3.44	2.42	9.5	9.17
Global moments	Data		Model	
GDP ratio (US/China) 2015	1.61		1.54	
Temperature rise by 2100 (°C)	2		2.01	

6 Results

6.1 Closed Economy

I consider a policy where the planner can only implement a constant carbon tax as from the year 2025. The social planner chooses a tax rate τ^d in all periods to maximise a utilitarian social welfare function, subject to the individual resource constraints (41), innovation decisions of firms (18) and (19), and the general equilibrium effects on wages (39). Under this restriction of time-invariant policies, the optimal carbon tax is 12%. Figure 3 shows green transition with and without the tax.

Figure 3. Effects of carbon taxes (Closed economy)

Panel 1: Change in GDP growth rate, share of dirty firms, clean R&D intensity and emissions relative to the no-tax baseline

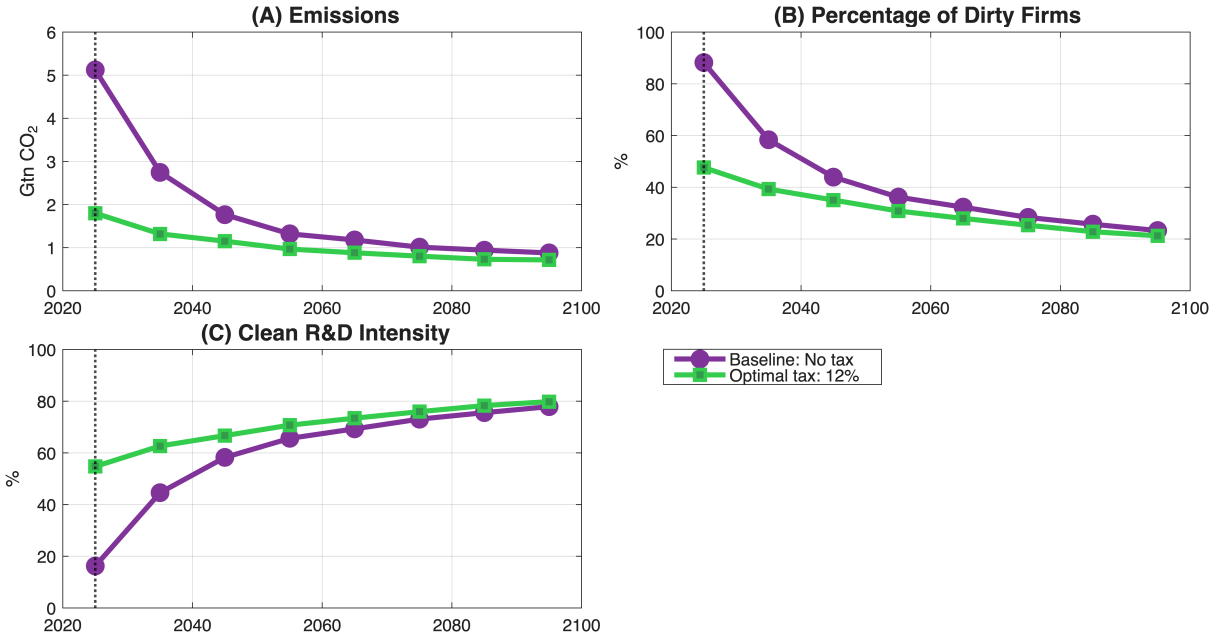


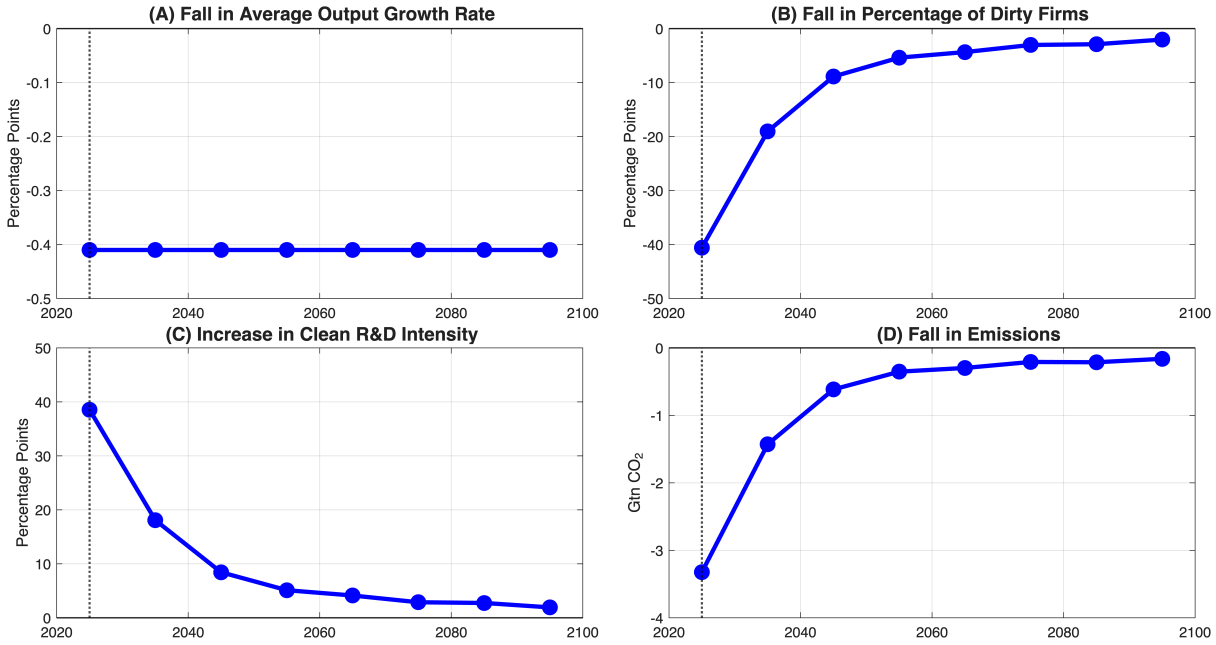
Figure 4 shows the changes economic and environmental variable relative to the no-tax baseline. The first graph in Panel 1 presents the average output growth rates. The introduction of a 12% tax leads to 0.4 percentage point lower output growth. This reflects the inefficiencies created by the carbon tax. The policy distorts the technology choices of firms and it is not necessarily the most productive firm that produces. This output loss is transmitted through two channels. First, the tax raises the cost of production of dirty firms, and hence reduces output. Second, wages adjust endogenously to productivity, and as a result grow at a slower rate relative to the baseline economy. The second graph in Panel 1 shows the evolution of the share of dirty firms in the economy, which decreases by 40 percentage points in 2025 relative to the baseline, and keeps falling over time. The sharp drops between 2025 and 2035 is driven by the fact that a large fraction of sectors start with productivity gaps clustered near the threshold, as shown in the initial distribution of technology gaps in Figure 2. A carbon tax accelerates the transition to clean technology because the tax shifts the tax-adjusted technology gap threshold

$$m_t = \frac{1}{ln\,\lambda}\left[ln\Big((1+\tau_t^d)\tilde{P}_t\Big)\right]$$

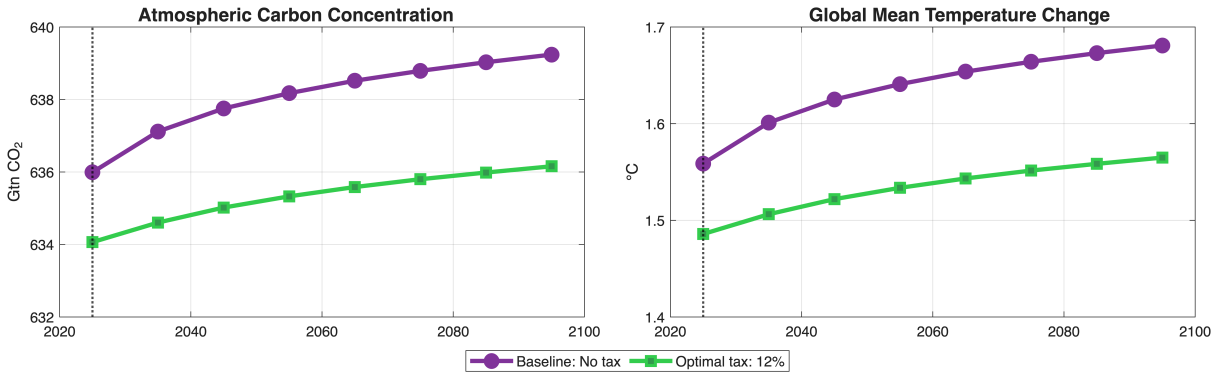
As explained in Lemma 3, a clean firm operates in sector j if $n_{j,t}^d - n_{j,t}^c \leq m_t$. A higher tax rate τ_t^d increases m_t and hence a higher share of sectors are clean.

Figure 4. Effects of carbon taxes (Closed economy)

Panel 1: Change in GDP growth rate, share of dirty firms, clean R&D intensity and emissions relative to the no-tax baseline



Panel 2: Environmental Outcomes



The third graph in Panel 1 shows the per-period change in clean R&D intensity, that is, how much of total R&D moves from dirty to clean relative to the no-tax economy. In 2025, 40% of R&D activity shifts from dirty firms to clean firms, compared with the baseline. As R&D increasingly shifts toward clean firms over time, the additional reallocation from dirty to clean R&D becomes progressively smaller.

The final graph in Panel 1 show the effects on emissions. Under the carbon tax, emissions fall by around 3.5Gtn in 2025. There are two margins for the reduction in emissions. First, the tax raises the cost of dirty production, reducing the quantity of the polluting fossil-fuel inputs demanded by dirty firms. Second, the tax accelerates the transition from

dirty to clean technologies through its effect on technology adoption decisions. Panel 2 shows the projections for atmospheric carbon concentration and the mean temperature changes relative to pre-industrial levels. In the baseline economy, atmospheric carbon concentration reaches approximately 640tn by 2100 corresponding to a temperature increase of almost 1.7°C. With a 12% carbon tax, carbon concentration increases to 636Gtn and temperature increases by 1.55°C by 2100.

6.2 Symmetric Countries

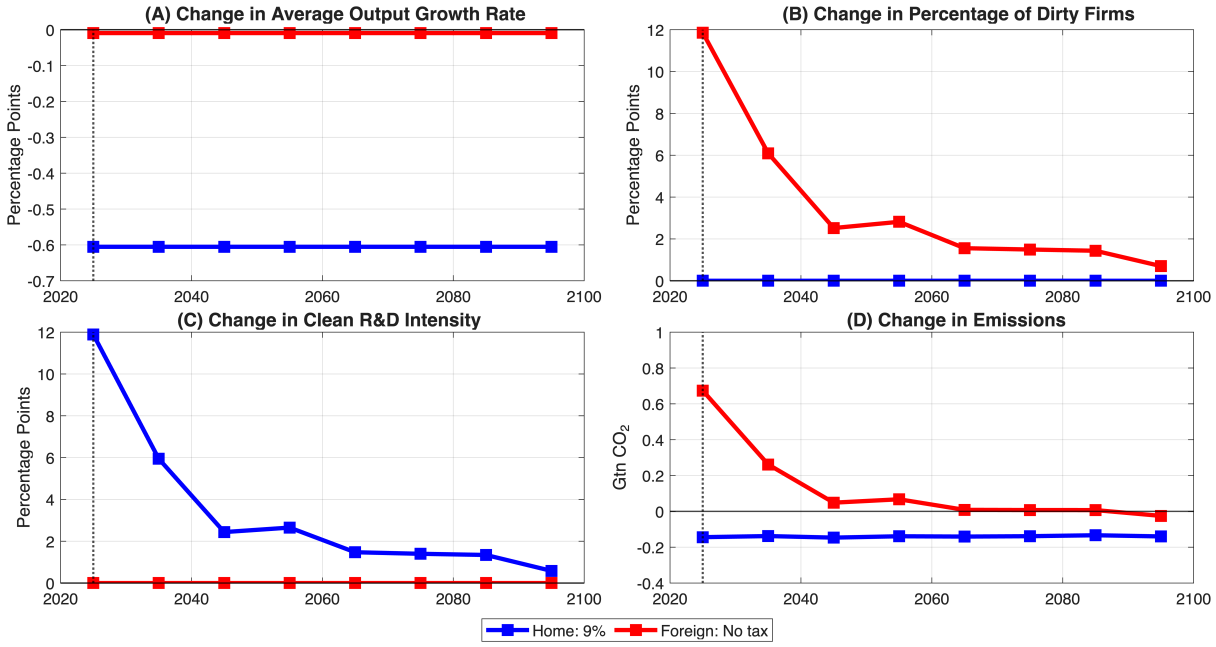
This section solves for the optimal policy under non-cooperation and cooperation. Both countries are symmetric ex-ante, with identical initial technology distributions. The home-bias parameter is set to $\epsilon = 0.52$, allowing for moderate substitutability between domestic and foreign intermediates. The global optimal carbon tax maximises global welfare with equal weights on each country. Under global cooperation, the optimal tax rate is 9% as leakage disappears and innovation becomes viable. To compare the effectiveness of a global tax with the unilateral case, I introduce a Home carbon tax in the year 2025, set at the global optimal rate of 9%, while Foreign maintains zero taxation throughout. Figure 5 shows the average output growth rate, share of dirty firms and emissions in the two countries before and after taxes.

Panel 1 present the main economic outcomes. The first graph shows that the average GDP growth rate of Home declines by 0.6 percentage points following the introduction of the tax. This is explained by the higher production costs of Home dirty intermediates, reducing competitiveness and shifting the demand towards Foreign intermediates. While the carbon tax generates within-country productivity advantages for some Home clean firms relative to their domestic dirty counterparts, cross-country comparisons shows that the Home clean production is still competitively dominated by untaxed Foreign dirty production. This explains the shift in demand towards Foreign intermediates. In contrast, Foreign's average growth rate remains almost unchanged, suggesting that the untaxed country captures a higher market share of intermediate goods without incurring any policy costs.

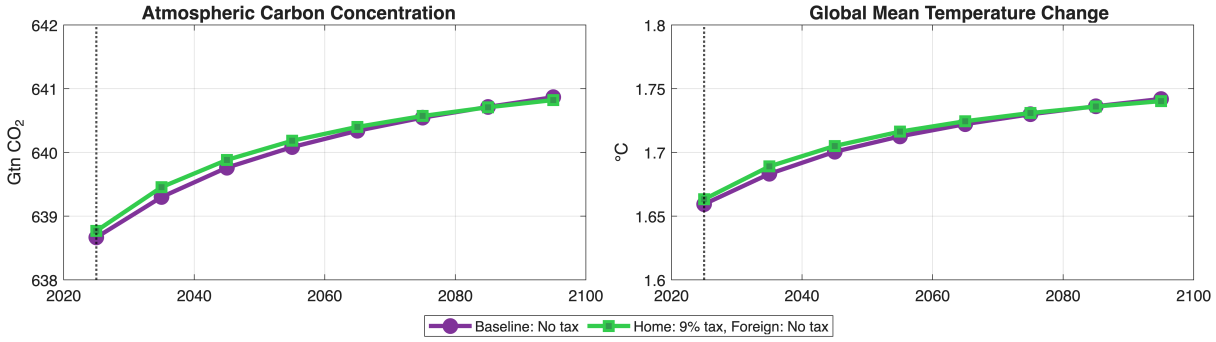
The second and third graphs show that while Home clean R&D share increases, the share of dirty firms remains largely unchanged post-tax. On the other hand, the clean R&D share in Foreign remains unchanged, and the share of dirty firms increases relative to the pre-tax economy, indicating an expansion in polluting production in the untaxed jurisdiction. Despite implementing a carbon tax, Home achieves minimal emissions reductions, as Home emissions fall by around 0.2Gtn, while Foreign emissions increases by 0.6Gtn, as shown in the fourth graph. Consequently, atmospheric carbon concentration and global

Figure 5. Effects of carbon taxes (Symmetric countries)

Panel 1: Change in GDP growth rate, share of dirty firms, clean R&D intensity and emissions relative to the no-tax baseline



Panel 2: Global environmental outcomes



average temperature change remain indistinguishable from the no-tax baseline, as shown in Panel 2, with temperature rising to 1.75°C above pre-industrial levels in both cases by 2100. Figure 6 shows the leakage rate of emissions which is calculated as ratio of the change in foreign emissions to the change in domestic emissions

$$L_t = -\frac{\Delta E_t^*}{\Delta E_t}$$

A leakage rate $L = 0$ means that there are no leakages and global emissions falls by the same amount as the reductions in Home emissions. When $0 < L < 1$, this indicates partial leakage, i.e., a reduction in global emissions, but less than the reduction in Home

emissions. $L = 1$ means that the Home reduction in emissions entirely by higher emissions in Foreign, while $L > 1$ shows negative spillovers of the policy and an increase in global emissions.

Figure 6. Leakage rate

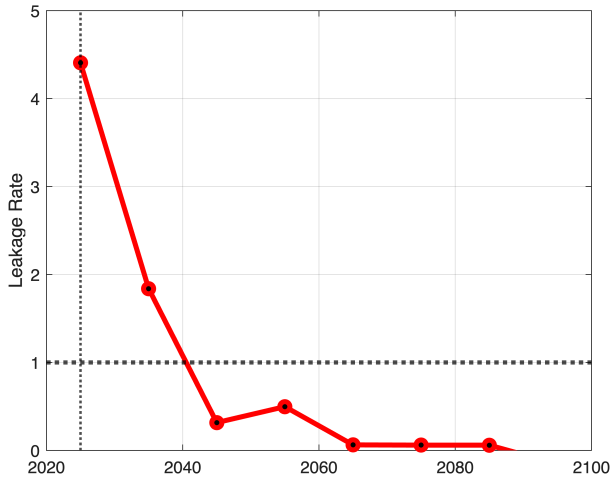


Figure 6 shows that Foreign emissions increases by a factor of 4.5 and 2 relative to Home reduction immediately after the introduction of the tax and the period after, respectively. This means that the unilateral policy actually increases global emissions initially. The competitiveness channel drives this result. The tax immediately increases production costs in Home, contracting output sharply, and Foreign firms expand to fill this market gap, increasing emissions. However, from 2045 onwards, the leakage rate is between 0 and 1 (partial leakage), implying that there is a reduction in global emissions, but less than the reduction in Home.

The welfare consequences for Home are unambiguously negative with the unilateral tax. Home incurs lower growth rate while achieving negligible climate benefits. Concurrently, Foreign experiences a gain in welfare from the increased market share without bearing any policy costs, which illustrates the free-rider problem associated with climate policies. The ineffectiveness of the policy arise from two mechanisms in the model. First, in the quality ladder framework, firms must invest in R&D to shift from dirty to clean production. A single country's tax generates insufficient R&D incentive, particularly when the untaxed Foreign market remains accessible. Second, the trade channel allows emissions to relocate rather than disappear. As Home reduces dirty production, Foreign expands to serve both markets. This substitution is almost one-for-one given the symmetric technology and absence of trade costs. Unlike a closed economy where domestic policy directly constrains all production, the open economy setting allows for the pollution-haven effect.

These findings rationalise the deadlocks in international climate negotiations and the

difficulty in sustaining ambitious unilateral policies. Countries implementing carbon taxes face competitiveness losses while delivering minimal climate benefits because of the pollution-haven effect. The results also speak to the current policy debate on border carbon adjustments (BCA). By imposing tariffs on imports from untaxed jurisdictions, then BCA policies can potentially tackle the leakage channel.

6.2.1 Policy Effectiveness

Figure 7 compares effectiveness of a unilateral tax and a global tax of 9%. Despite implementing the same tax, the outcomes in the two economies contrast sharply. Panel A shows that output growth rate falls by 0.7 percentage points in the global-tax economy and 0.6 percentage points in the unilateral-tax economy. While the open economy faces lower economic costs, the changes in the share of dirty firms and emissions (second and third graphs) with a unilateral tax are negligible relative to the no-tax baseline. The slight drop in emissions happens because of lower production and not because of clean transition. On the other hand, in the global-tax economy, the environmental impacts of the policy are significant. The share of dirty firms drops by 10 percentage points immediately after the introduction of the tax and emissions fall by approximately 1.2Gtn.

Panel 2 shows the share of clean R&D. In the global-tax economy, the carbon tax induces an increase of R&D toward clean technology from 60% in 2025 immediately after the introduction of the tax to 80% by 2100. By contrast, the share of clean R&D decreases over time in the unilateral-tax economy from 50% in 2025 to 25% in 2100. This shows that without coordinated carbon taxes, firms have less incentives to invest in clean R&D. In the global-tax economy, the tax redirects R&D from dirty to clean technologies, while in the unilateral-tax case, it has a lower impact on innovation incentives as domestic firms have to compete internationally with untaxed cheaper dirty alternatives. The innovation channel, emphasised by Acemoglu et al. (2012, 2016) and Aghion et al. (2016) as the main mechanism for long-run decarbonisation, is offset by international trade and leakages.

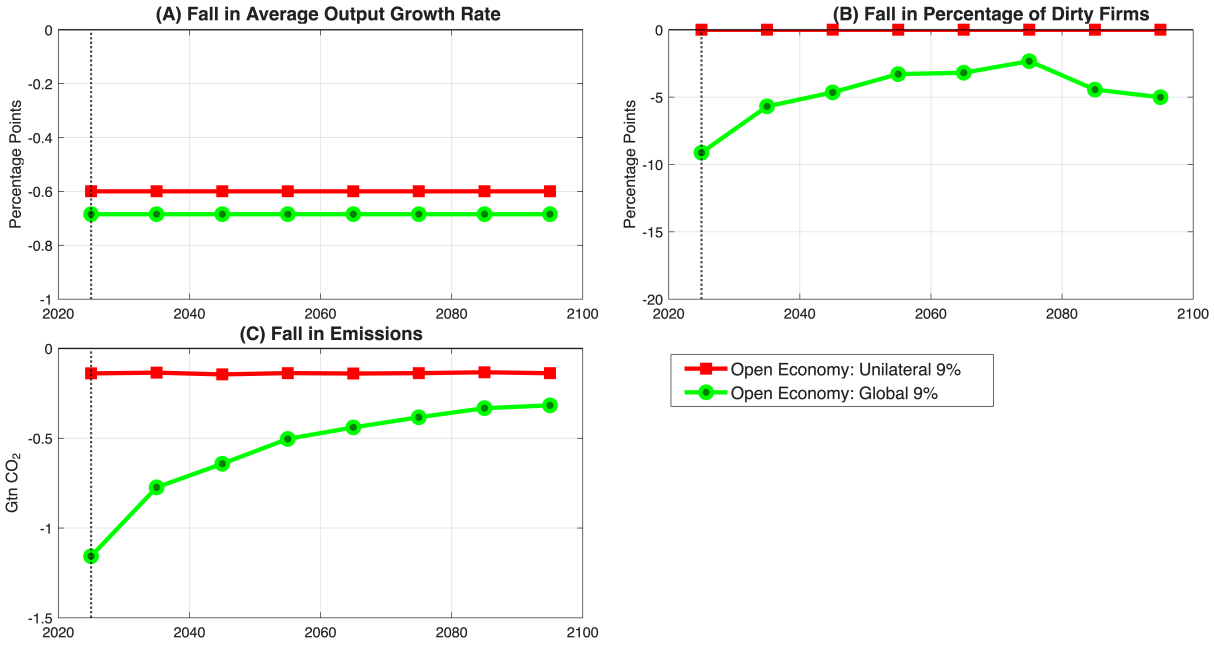
6.3 Asymmetric Countries

The social planner chooses time-invariant tax rates (τ_H^d, τ_F^d), introduced in 2025, to maximise a utilitarian social welfare function, subject to the resource constraint, optimal innovation decisions of firms and the general equilibrium effects on wages. The optimal carbon tax is 6% for the US and 30% for China. Figure 8 shows the economic and environmental effects of the optimal taxes relative to the no-tax baseline economy.

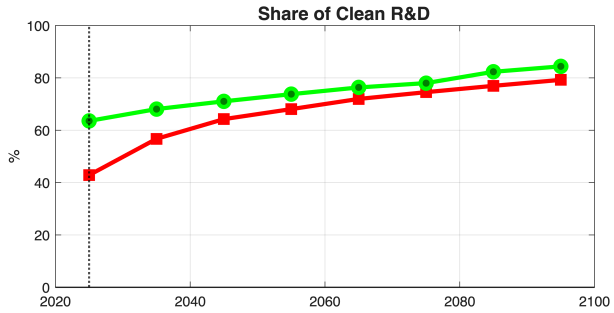
The first graph in Panel 1 shows that the average GDP growth rate declines by 0.05

Figure 7. Effectiveness of a unilateral and global tax

Panel 1: Change in GDP growth rate, share of dirty firms and emissions relative to the no-tax baseline



Panel 2: R&D

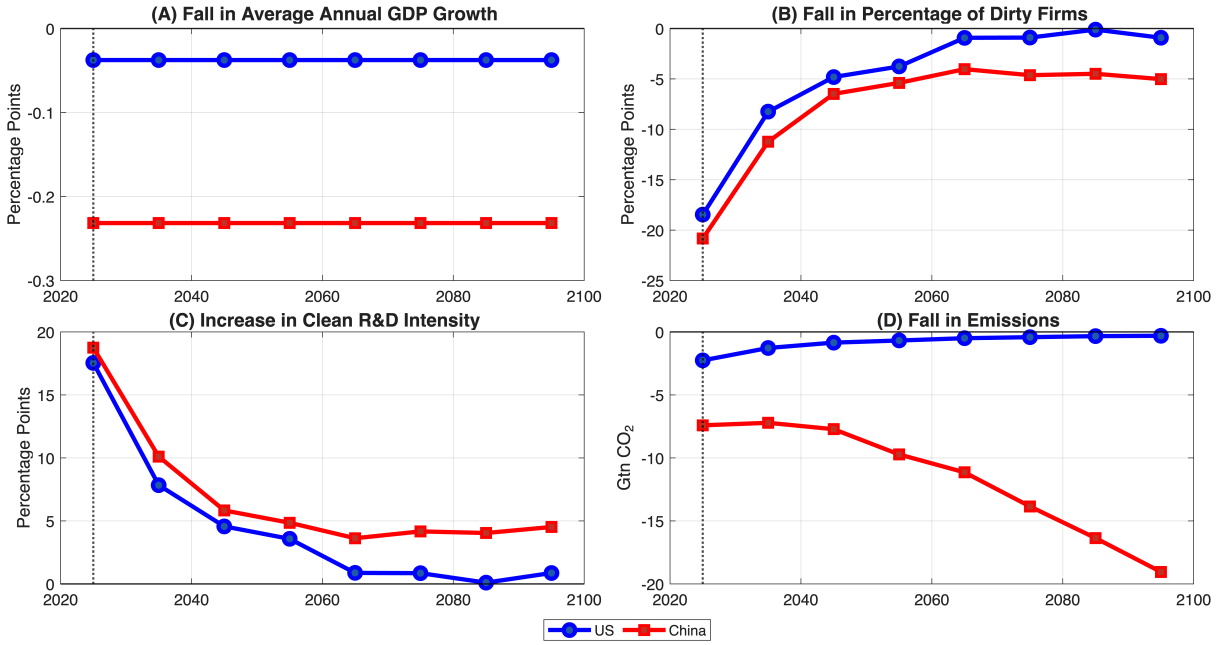


percentage points in the US and 0.25 percentage points in China. The second and third graphs show that the percentage of dirty firms fall, and there is a switch in R&D activities from dirty to clean firms in both countries. In the fourth graph, emissions fall by approximately 3Gtn in the US and 7Gtn in China upon the introduction of the tax. While the fall in emissions becomes gradually smaller in the US, in China, emissions falls by a greater amount until 2100. This means that absent the tax, emissions in China would have increased continuously over time.

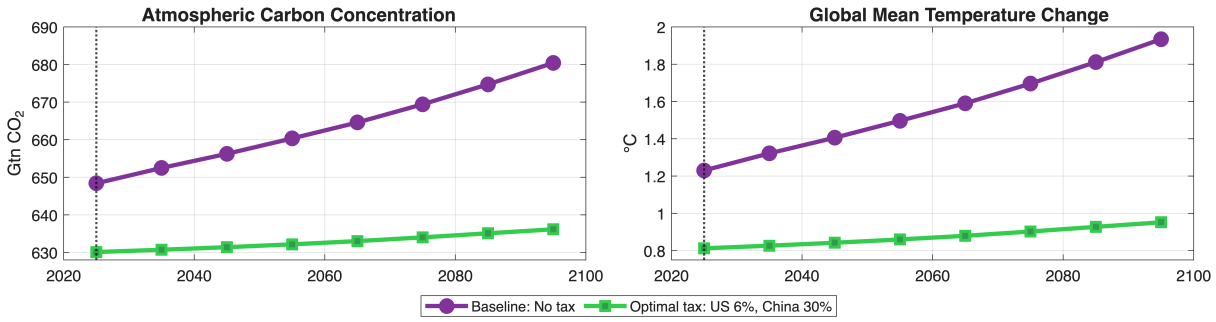
Panel 2 shows the evolution of atmospheric carbon concentration and average temperature changes relative to pre-industrial levels. Without taxes, atmospheric carbon rises to 680Gtn by 2100, driving a temperature increase of almost 2°C. Introducing the optimal

Figure 8. Effects of carbon taxes (Asymmetric countries)

Panel 1: Change in GDP growth rate, share of dirty firms, clean R&D intensity and emissions relative to the no-tax baseline



Panel 2: Global environmental outcomes



taxes significantly slows the accumulation of atmospheric carbon which peaks at 638Gtn, representing a reduction of around 6%, while the temperature change is limited to 1°C.

6.3.1 Nash Equilibrium Taxes

Having characterised the global optimal tax rate, I now consider the case where Home and Foreign set their carbon taxes unilaterally, each maximizing its own welfare taking the other country's tax rate as given. The trade and leakage channels ties the two countries' production, innovation and emissions together, generating strategic interactions such that each country's optimal unilateral tax depends on the tax of the other. I characterise the equilibrium tax as a one-shot, simultaneous move game and solve for its Nash equilibrium.

Each country's best response is, respectively

$$BR^H\big((\tau^d)^*\big) \;=\; \arg\max_{\tau^d\in[\underline{\tau},\overline{\tau}]} W^H \qquad\qquad BR^F\big(\tau^d\big) \;=\; \arg\max_{(\tau^d)^*\in[\underline{\tau},\overline{\tau}]} W^F$$

and a Nash equilibrium is a pair $\{\tilde{\tau^d}, (\tau^{\tilde{d}})^*\}$ satisfying mutual best responses

$$\tilde{\tau^d} \;=\; BR^H\big((\tau^{\tilde{d}})^*\big) \qquad\qquad (\tau^{\tilde{d}})^* \;=\; BR^F(\tilde{\tau^d})$$

A grid search for the optimal policy reveals that there is no positive welfare-improving unilateral tax rate. The main mechanism driving this result is the asymmetry between the costs and benefits of unilateral actions. Any positive carbon tax generates domestic economic costs through output contraction as dirty production becomes more expensive and competitiveness losses as production relocates abroad. As shown in the previous section, carbon leakage ensures that any reductions in Home emissions are largely offset by Foreign increases, and any potential environmental benefits are negligible.

The political economy implications are twofold. First, a rational government facing domestic welfare maximisation will not implement meaningful climate policies absent credible commitments from other countries. Unilateral actions in open economies are not simply "less effective", but instead, they are welfare-reducing relative to inaction. Second, while some advocates argue that early adopters of climate policies benefit from induced innovation in clean technologies (Porter and Linde, 1995; Fankhauser et al., 2013), this framework demonstrates that innovation requires clean technology to be profitable, which is not the case when firms must compete against untaxed Foreign dirty production.

7 Time-Varying Taxes

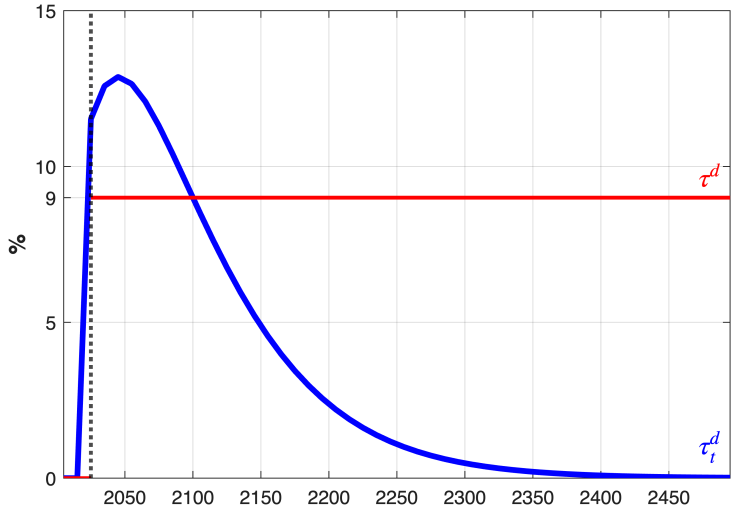
In the previous section, I assume that the planner chooses a constant carbon tax rate over time. While this assumption is not implausible given how taxes are introduced in practice, it is also informative to study an environment in which the planner commits to a time-varying sequence of carbon taxes. To characterise the optimal time-dependent tax in the symmetric case, I parameterise the tax path as

$$\tau_t^d \;=\; \overline{\tau} t e^{-\xi t} \tag{47}$$

This function starts at zero in the year 2015, rises to a peak of $\frac{\overline{\tau}}{\xi e}$ at time $t = 1/\xi$, and then decreases toward zero as the transition to clean technology is completed. ξ controls the timing of the peak, while $\overline{\tau}$ controls the scale of the tax path. This parametric approach

keeps the parameter space low-dimensional and computationally tractable. I search over a grid of $(\overline{\tau}, \xi)$ values and select the pair that maximises discounted welfare. The optimal carbon taxes in the symmetric are shown in Figure 9.

Figure 9. Optimal time-varying taxes (Symmetric countries)



The optimal carbon tax is hump-shaped. It starts out at 11% in 2025. increases to a peak of 12.5% over the next 20 years and then decreases gradually back down to zero after 300 years, the time by which the clean transition is complete. The welfare gains with time-varying taxes are 2 percentage points higher relative to time-invariant taxes.

8 Subsidies on Clean R&D

This section introduces a research subsidy χ as well as a carbon tax τ^d on dirty firms, chosen unilaterally by the Home country to maximise welfare. Clean firms now account for the subsidy when hiring scientists and their value functions (19) are given by

$$\begin{aligned} V^{c,i}_{j,t}(n^d_{j,t} - n^c_{j,t} - m_t) \;=\; \max_{s^{c,i}_{j,t} \geq 0} \mathbb{1}_{\{n^d_{j,t} - n^c_{j,t} \leq m_t\}} \pi^c_{j,t} \;-\; (1-\chi) w^s_t s^{c,i}_{j,t} \\ +\; \rho \mathbb{E} V^{c,i}_{j,t+1}(n^d_{j,t+1} - n^c_{j,t+1} - m_{t+1}) \end{aligned} \tag{48}$$

9 Conclusion

This paper studies how international coordination changes the interaction between carbon taxation and directed technological change. I develop a two-country dynamic general equilibrium model in which clean and dirty firms compete within each sector, firms direct R&D toward the two technologies, and dirty production generates carbon emissions

that contribute to a common global carbon stock. Carbon taxation affects both current production and the endogenous evolution of technology. The closed-economy analysis illustrates the standard directed technical change mechanism. By increasing the relative cost of dirty production, a carbon tax shifts production toward clean firms and raises the expected return to clean innovation. Given that a substantial mass of sectors lies close to the clean-dirty technology threshold, this response can be rapid. The optimal time-invariant policy produces a large initial reallocation of production and R&D toward clean technologies and significantly lowers emissions.

In the two-country case, a unilateral carbon tax reduces the competitiveness of domestic dirty producers, causing production to shift toward untaxed foreign firms. The resulting carbon leakage offsets the domestic reduction in emissions and weakens the incentives for clean innovation. In the calibrated symmetric economy, foreign emissions initially increase by more than the reduction in domestic emissions, leaving global climate outcomes almost unchanged. Consequently, a government maximising domestic welfare has little incentives to implement unilateral policies, whereas a coordinated carbon tax restores both the emissions and innovation channels.

In the US-China calibration, I show that international asymmetries also matter for the optimal policy design even under a common global climate externality. Thus, coordination does not imply uniform carbon prices when countries differ in their productive and technological structures. These findings emphasise that the effectiveness of environmental policies depends crucially on both the domestic response of firms and international coordination. Policies that increase the cost of dirty production in one jurisdiction can redirect production rather than eliminate it, thereby weakening the long-run technological response that makes carbon taxation effective in a closed economy. This provides a rationale for international policy coordination and for instruments, such as border carbon adjustments, that directly address the leakage margin.

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Appendix

Proof of Lemma 1

The exhaustible resource owner solves the following problem

$$\max_{\{q_t^e\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \left(\prod_{s=0}^{s-1} \frac{1}{1+r_{s+1}^w} \right) p_t^e q_t^e \qquad \text{s.t} \qquad \sum_{t=0}^{\infty} q_t^e \;=\; R_0$$

The first order condition of the Lagrangian is

$$\frac{\partial L}{\partial q_t^e} \;=\; \left(\prod_{s=0}^{t-1} \frac{1}{1+r_{s+1}^w} \right) p_t^e \;-\; \lambda \;=\; 0$$

For two consecutive periods

$$\left(\prod_{s=0}^{t-1} \frac{1}{1+r_{s+1}^w} \right) p_t^e \;=\; \left(\prod_{s=0}^{t-2} \frac{1}{1+r_{s+1}^w} \right) p_{t-1}^e$$

$$p_t^e \;=\; (1+r_t^w) p_{t-1}^e$$

Replace recursively to complete the proof

$$p_t^e \;=\; \left(\prod_{s=1}^{t} (1+r_s^w) \right) p_0^e$$

Proof of Lemma 2

The cost minimisation problem of clean producers is

$$\min_{l_{j,t}^c} \; w_t^u l_{j,t}^c \quad \text{s.t} \quad \theta_{j,t}^c l_{j,t}^c \;=\; y_{j,t}^c$$

giving a marginal cost of

$$\frac{w_t^u}{\theta_{j,t}^c} \tag{A.1}$$

The cost minimisation problem of dirty producers (dropping the sector and time indices for simplicity) is

$$\min_{\{l^d, x\}} \; (1+\tau^d)(w^u l^d \;+\; p^e e) \quad \text{s.t} \quad \theta^d (l^d)^{1-\gamma} e^{\gamma} \;=\; y^d \tag{A.2}$$

The Lagrangian is

$$L \;=\; (1+\tau^d)(w^u l^d \;+\; p^e e) \;-\; \lambda\Big[\theta^d (l^d)^{1-\gamma} e^{\gamma} \;-\; y^d\Big]$$

The first order conditions are

$$(l^d): \qquad (1+\tau^d)w^u \;-\; \lambda\Big[\theta^d (l^d)^{-\gamma} e^{\gamma}(1-\gamma)\Big] \;=\; 0 \tag{A.3}$$

$$(e): \qquad (1+\tau^d)p^e \;-\; \lambda\Big[\theta^d (l^d)^{1-\gamma} e^{\gamma-1}\gamma\Big] \;=\; 0 \tag{A.4}$$

Solving this gives the optimal input demands

$$l^d \;=\; \left[\frac{p^e}{w^u}\frac{(1-\gamma)}{\gamma}\right]^{\gamma}\frac{y^d}{\theta^d} \qquad\qquad e \;=\; \left[\frac{w^u}{p^e}\frac{(\gamma)}{1-\gamma}\right]^{1-\gamma}\frac{y^d}{\theta^d}$$

Substituting the optimal inputs, the total cost is given by

$$(1+\tau^d)(w^u l^d \;+\; p^e e) \;=\; (1+\tau^d)\frac{(w^u)^{1-\gamma}(p^e)^{\gamma}}{\gamma^{\gamma}(1-\gamma)^{1-\gamma}}\frac{y^d}{\theta^d}$$

Finally, the marginal cost is

$$\frac{(1+\tau^d)w^u\tilde{P}}{\theta^d} \qquad \text{where} \qquad \tilde{P} \;=\; \left[\frac{p^e}{w^u\gamma}\right]^{\gamma}(1-\gamma)^{\gamma-1} \tag{A.5}$$

Compare (A.1) and (A.5) to get Lemma 2.

Proof of Lemma 3

The technology threshold can be written as

$$\text{Clean technology if} \qquad 1 \;>\; \frac{1}{(1+\tau_t^d)\tilde{P}_t}\lambda^{n^d_{j,t}-n^c_{j,t}}$$

$$\text{Dirty technology if} \qquad 1 \;<\; \frac{1}{(1+\tau_t^d)\tilde{P}_t}\lambda^{n^d_{j,t}-n^c_{j,t}}$$

Taking logs and re-arranging to complete the proof

$$\text{Clean technology if} \qquad n^d_{j,t} - n^c_{j,t} \;<\; m_t$$

$$\text{Dirty technology if} \qquad n^d_{j,t} - n^c_{j,t} \;>\; m_{j_t}$$

where

$$m_t \;=\; \frac{1}{ln\,\lambda}\left[ln\Big((1+\tau_t^d)\tilde{P}_t\Big)\right] \tag{A.6}$$

is the price-adjusted policy gap in each sector.

Proof of Corollary 1 Re-writing (17) as

$$\frac{\theta_{j,t}^d}{\theta_{j,t}^c}\lambda^{-m_t} \;=\; \lambda^{n_{j,t}^d - n_{j,t}^c - m_t}$$

From (A.6)

$$\begin{aligned}
m_t &= \frac{1}{ln\,\lambda}\left[ln\Big((1+\tau_t^d)\tilde{P}_t\Big)\right] \\
m_t ln\,\lambda &= ln\Big((1+\tau_t^d)\tilde{P}_t\Big) \\
-m_t ln\,\lambda &= -ln\Big((1+\tau_t^d)\tilde{P}_t\Big) \\
ln\,\lambda^{-m_t} &= ln\Big((1+\tau_t^d)\tilde{P}_t\Big)^{-1} \\
\lambda^{-m_t} &= \Big((1+\tau_t^d)\tilde{P}_t\Big)^{-1} \;=\; \frac{1}{1+\tau_t^d}\frac{1}{\tilde{P}_t}
\end{aligned}$$

Substitute back in the first expression to complete the proof.

Proof of Proposition 1

Since $(\hat{\tau}_H^d, \hat{\tau}_F^d)$ is an interior solution, the planner's first-order condition with respect to country m's tax rate is

$$\left.\frac{\partial W_m(\tau_m^d, \tau_{-m}^d)}{\partial \tau_m^d}\right|_{(\hat{\tau}_H^d, \hat{\tau}_F^d)} + \left.\frac{\partial W_{-m}(\tau_m^d, \tau_{-m}^d)}{\partial \tau_m^d}\right|_{(\hat{\tau}_H^d, \hat{\tau}_F^d)} = 0$$

By assumption,

$$\left.\frac{\partial W_{-m}(\tau_m^d, \tau_{-m}^d)}{\partial \tau_m^d}\right|_{(\hat{\tau}_H^d, \hat{\tau}_F^d)} \neq 0$$

Therefore,

$$\left.\frac{\partial W_m(\tau_m^d, \tau_{-m}^d)}{\partial \tau_m^d}\right|_{(\hat{\tau}_H^d, \hat{\tau}_F^d)} = -\left.\frac{\partial W_{-m}(\tau_m^d, \tau_{-m}^d)}{\partial \tau_m^d}\right|_{(\hat{\tau}_H^d, \hat{\tau}_F^d)} \neq 0.$$

Hence $\tau_m^d = \hat{\tau}_m^d$ cannot be an interior solution of $W_m(\tau_m^d, \hat{\tau}_{-m}^d)$. Since $\hat{\tau}_m^d$ is interior,

there exists a feasible local deviation from $\hat{\tau}_m^d$ that strictly increases country m's welfare. Therefore

$$\hat{\tau}_m^d \notin BR_m(\tau_{-m}^d = \hat{\tau}_{-m}^d).$$

It follows that $(\hat{\tau}_H^d, \hat{\tau}_F^d)$ is not a Nash equilibrium.

When

$$\left.\frac{\partial W_{-m}(\tau_m^d, \tau_{-m}^d)}{\partial \tau_m^d}\right|_{(\hat{\tau}_H^d, \hat{\tau}_F^d)} > 0.$$

It follows that,

$$\left.\frac{\partial W_m(\tau_m^d, \tau_{-m}^d)}{\partial \tau_m^d}\right|_{(\hat{\tau}_H^d, \hat{\tau}_F^d)} < 0.$$

Thus, holding country $-m$'s tax rate fixed at $\hat{\tau}_{-m}^d$, country m's welfare is locally decreasing in its own tax rate at $\hat{\tau}_m^d$. Since the cooperative solution is interior, a small reduction in country m's dirty tax rate is feasible, and for a sufficiently small $\varepsilon > 0$,

$$W_m(\hat{\tau}_m^d - \varepsilon, \hat{\tau}_{-m}^d) > W_m(\hat{\tau}_m^d, \hat{\tau}_{-m}^d).$$